

1 *Quality Improvement in the Modern Business Environment*

CHAPTER OUTLINE

1.1	THE MEANING OF QUALITY AND QUALITY IMPROVEMENT	1.4	MANAGEMENT ASPECTS OF QUALITY IMPROVEMENT
1.1.1	Dimensions of Quality	1.4.1	Quality Philosophy and Management Strategies
1.1.2	Quality Engineering Terminology	1.4.2	The Link Between Quality and Productivity
1.2	A BRIEF HISTORY OF QUALITY CONTROL AND IMPROVEMENT	1.4.3	Quality Costs
1.3	STATISTICAL METHODS FOR QUALITY CONTROL AND IMPROVEMENT	1.4.4	Legal Aspects of Quality
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CHAPTER OVERVIEW AND LEARNING OBJECTIVES

This book is about the use of statistical methods and other problem-solving techniques to improve the **quality** of the products used by our society. These products consist of **manufactured goods** such as automobiles, computers, and clothing, as well as **services** such as the generation and distribution of electrical energy, public transportation, banking, retailing, and health care. Quality improvement methods can be applied to any area within a company or organization, including manufacturing, process development, engineering design, finance and accounting, marketing, distribution and logistics, customer service, and field service of products. This text presents the technical tools that are needed to achieve quality improvement in these organizations.

In this chapter we give the basic definitions of quality, quality improvement, and other quality engineering terminology. We also discuss the historical development of quality

improvement methodology and overview the statistical tools essential for modern professional practice. A brief discussion of some management and business aspects for implementing quality improvement is also given.

After careful study of this chapter you should be able to do the following:

1. Define and discuss quality and quality improvement
2. Discuss the different dimensions of quality
3. Discuss the evolution of modern quality improvement methods
4. Discuss the role that variability and statistical methods play in controlling and improving quality
5. Describe the quality management philosophies of W. Edwards Deming, Joseph M. Juran, and Armand V. Feigenbaum
6. Discuss total quality management, the Malcolm Baldrige National Quality Award, six-sigma, and quality systems and standards
7. Explain the links between quality and productivity and between quality and cost
8. Discuss product liability
9. Discuss the three functions: quality planning, quality assurance, and quality control and improvement

1.1 The Meaning of Quality and Quality Improvement

We may define **quality** in many ways. Most people have a conceptual understanding of quality as relating to one or more desirable characteristics that a product or service should possess. Although this conceptual understanding is certainly a useful starting point, we will give a more precise and useful definition.

Quality has become one of the most important consumer decision factors in the selection among competing products and services. The phenomenon is widespread, regardless of whether the consumer is an individual, an industrial organization, a retail store, a bank or financial institution, or a military defense program. Consequently, understanding and improving quality are key factors leading to business success, growth, and enhanced competitiveness. There is a substantial return on investment from improved quality and from successfully employing quality as an integral part of overall business strategy. In this section we provide operational definitions of quality and quality improvement. We begin with a brief discussion of the different dimensions of quality and some basic terminology.

1.1.1 Dimensions of Quality

The quality of a product can be described and evaluated in several ways. It is often very important to differentiate these different **dimensions of quality**. Garvin (1987) provides an excellent discussion of eight components or dimensions of quality. We summarize his key points concerning these dimensions of quality as follows:

1. **Performance** (Will the product do the intended job?) Potential customers usually evaluate a product to determine if it will perform certain specific functions and determine how well it performs them. For example, you could evaluate spreadsheet software packages for a PC to determine which data manipulation operations they perform. You may discover that one outperforms another with respect to the execution speed.

2. **Reliability** (How often does the product fail?) Complex products, such as many appliances, automobiles, or airplanes, will usually require some repair over their service life. For example, you should expect that an automobile will require occasional repair, but if the car requires frequent repair, we say that it is unreliable. There are many industries in which the customer's view of quality is greatly impacted by the reliability dimension of quality.
3. **Durability** (How long does the product last?) This is the effective service life of the product. Customers obviously want products that perform satisfactorily over a long period of time. The automobile and major appliance industries are examples of businesses where this dimension of quality is very important to most customers.
4. **Serviceability** (How easy is it to repair the product?) There are many industries in which the customer's view of quality is directly influenced by how quickly and economically a repair or routine maintenance activity can be accomplished. Examples include the appliance and automobile industries and many types of service industries (how long did it take a credit card company to correct an error in your bill?).
5. **Aesthetics** (What does the product look like?) This is the visual appeal of the product, often taking into account factors such as style, color, shape, packaging alternatives, tactile characteristics, and other sensory features. For example, soft-drink beverage manufacturers have relied on the visual appeal of their packaging to differentiate their product from other competitors.
6. **Features** (What does the product do?) Usually, customers associate high quality with products that have added features; that is, those that have features beyond the basic performance of the competition. For example, you might consider a spreadsheet software package to be of superior quality if it had built-in statistical analysis features while its competitors did not.
7. **Perceived Quality** (What is the reputation of the company or its product?) In many cases, customers rely on the past reputation of the company concerning quality of its products. This reputation is directly influenced by failures of the product that are highly visible to the public or that require product recalls, and by how the customer is treated when a quality-related problem with the product is reported. Perceived quality, customer loyalty, and repeated business are closely interconnected. For example, if you make regular business trips using a particular airline, and the flight almost always arrives on time and the airline company does not lose or damage your luggage, you will probably prefer to fly on that carrier instead of its competitors.
8. **Conformance to Standards** (Is the product made exactly as the designer intended?) We usually think of a high-quality product as one that exactly meets the requirements placed on it. For example, how well does the hood fit on a new car? Is it perfectly flush with the fender height, and is the gap exactly the same on all sides? Manufactured parts that do not exactly meet the designer's requirements can cause significant quality problems when they are used as the components of a more complex assembly. An automobile consists of several thousand parts. If each one is just slightly too big or too small, many of the components will not fit together properly, and the vehicle (or its major subsystems) may not perform as the designer intended.

We see from the foregoing discussion that quality is indeed a multifaceted entity. Consequently, a simple answer to questions such as "What is quality?" or "What is quality improvement?" is not easy. The **traditional** definition of quality is based on the viewpoint that products and services must meet the requirements of those who use them.

Definition

Quality means fitness for use.

There are two general aspects of fitness for use: **quality of design** and **quality of conformance**. All goods and services are produced in various grades or levels of quality. These variations in grades or levels of quality are intentional, and, consequently, the appropriate technical term is quality of design. For example, all automobiles have as their basic objective providing safe transportation for the consumer. However, automobiles differ with respect to size, appointments, appearance, and performance. These differences are the result of intentional design differences among the types of automobiles. These design differences include the types of materials used in construction, specifications on the components, reliability obtained through engineering development of engines and drive trains, and other accessories or equipment.

The quality of conformance is how well the product conforms to the specifications required by the design. Quality of conformance is influenced by a number of factors, including the choice of manufacturing processes, the training and supervision of the workforce, the types of process controls, tests, and inspection activities that are employed, the extent to which these procedures are followed, and the motivation of the workforce to achieve quality.

Unfortunately, this definition has become associated more with the conformance aspect of quality than with design. This is in part due to the lack of formal education most designers and engineers receive in quality engineering methodology. This also leads to much less focus on the customer and more of a “conformance-to-specifications” approach to quality, regardless of whether the product, even when produced to standards, was actually “fit-for-use” by the customer. Also, there is still a widespread belief that quality is a problem that can be dealt with solely in manufacturing, or that the only way quality can be improved is by “gold-plating” the product.

We prefer a **modern** definition of quality:

Definition

Quality is inversely proportional to variability.

Note that this definition implies that if variability¹ in the important characteristics of a product decreases, the quality of the product increases.

As an example of the operational effectiveness of this definition, a few years ago, one of the automobile companies in the United States performed a comparative study of a transmission that was manufactured in a domestic plant and by a Japanese supplier. An analysis of warranty claims and repair costs indicated that there was a striking difference between the two sources of production, with the Japanese-produced transmission having much lower costs, as shown in Fig. 1.1. As part of the study to discover the cause of this difference in cost and performance, the company selected random samples of transmissions from each plant, disassembled them, and measured several critical quality characteristics.

¹We are referring to unwanted or harmful variability. There are situations in which variability is actually good. As my good friend Bob Hogg has pointed out, “I really like Chinese food, but I don’t want to eat it every night.”

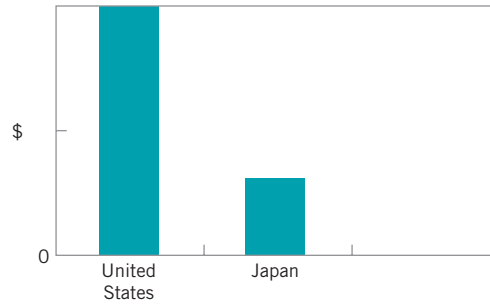


FIGURE 1.1 Warranty costs for transmissions.

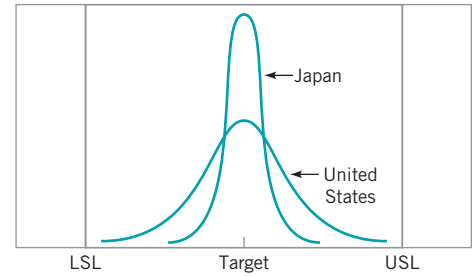


FIGURE 1.2 Distributions of critical dimensions for transmissions.

Figure 1.2 is generally representative of the results of this study. Note that both distributions of critical dimensions are centered at the desired or target value. However, the distribution of the critical characteristics for the transmissions manufactured in the United States takes up about 75% of the width of the specifications, implying that very few nonconforming units would be produced. In fact, the plant was producing at a quality level that was quite good, based on the generally accepted view of quality within the company. In contrast, the Japanese plant produced transmissions for which the same critical characteristics take up only about 25% of the specification band. As a result, there is considerably less variability in the critical quality characteristics of the Japanese-built transmissions in comparison to those built in the United States.

This is a very important finding. Jack Welch, the retired chief executive officer of General Electric, has observed that your customer doesn't see the mean of your process (the target in Fig. 1.2), he only sees the variability around that target that you have not removed. In almost all cases, this variability has significant customer impact.

There are two obvious questions here: Why did the Japanese do this? How did they do this? The answer to the "why" question is obvious from examination of Fig. 1.1. Reduced variability has directly translated into lower costs (the Japanese fully understood the point made by Welch). Furthermore, the Japanese-built transmissions shifted gears more smoothly, ran more quietly, and were generally perceived by the customer as superior to those built domestically. Fewer repairs and warranty claims means less **rework** and the reduction of wasted time, effort, and money. Thus, quality truly is inversely proportional to variability. Furthermore, it can be communicated very precisely in a language that everyone (particularly managers and executives) understands—namely, money.

How did the Japanese do this? The answer lies in the systematic and effective use of the methods described in this book. It also leads to the following definition of **quality improvement**.

Definition

Quality improvement is the reduction of variability in processes and products.

Excessive variability in process performance often results in **waste**. For example, consider the wasted money, time, and effort that is associated with the repairs represented in Fig. 1.1. Therefore, an alternate and frequently very useful definition is that quality improvement is the **reduction of waste**. This definition is particularly effective in **service industries**, where there may not be as many things that can be directly measured (like the transmission

critical dimensions in Fig. 1.2). In service industries, a quality problem may be an error or a mistake, the correction of which requires effort and expense. By improving the service process, this wasted effort and expense can be avoided.

We now present some quality engineering terminology that is used throughout the book.

1.1.2 Quality Engineering Terminology

Every product possesses a number of elements that jointly describe what the user or consumer thinks of as quality. These parameters are often called **quality characteristics**. Sometimes these are called **critical-to-quality (CTQ)** characteristics. Quality characteristics may be of several types:

1. **Physical:** length, weight, voltage, viscosity
2. **Sensory:** taste, appearance, color
3. **Time Orientation:** reliability, durability, serviceability

Note that the different types of quality characteristics can relate directly or indirectly to the dimensions of quality discussed in the previous section.

Quality engineering is the set of operational, managerial, and engineering activities that a company uses to ensure that the quality characteristics of a product are at the nominal or required levels and that the variability around these desired levels is minimum. The techniques discussed in the book form much of the basic methodology used by engineers and other technical professionals to achieve these goals.

Most organizations find it difficult (and expensive) to provide the customer with products that have quality characteristics that are always identical from unit to unit, or are at levels that match customer expectations. A major reason for this is **variability**. There is a certain amount of variability in every product; consequently, no two products are ever identical. For example, the thickness of the blades on a jet turbine engine impeller is not identical even on the same impeller. Blade thickness will also differ between impellers. If this variation in blade thickness is small, then it may have no impact on the customer. However, if the variation is large, then the customer may perceive the unit to be undesirable and unacceptable. Sources of this variability include differences in materials, differences in the performance and operation of the manufacturing equipment, and differences in the way the operators perform their tasks. This line of thinking led to the previous definition of quality improvement.

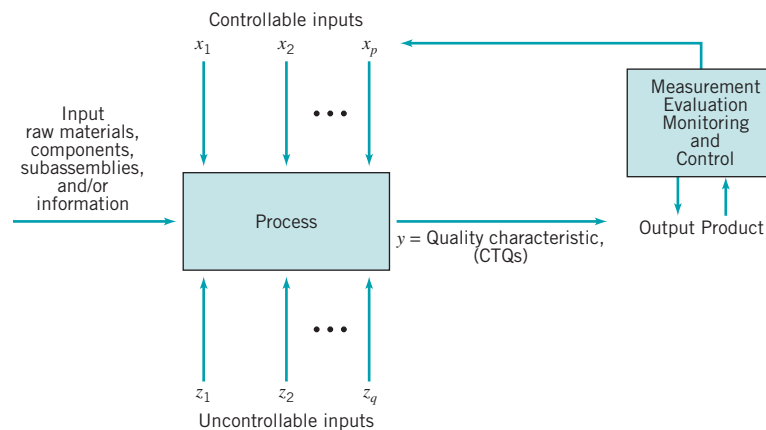
Since variability can only be described in statistical terms, **statistical methods** play a central role in quality improvement efforts. In the application of statistical methods to quality engineering, it is fairly typical to classify data on quality characteristics as either **attributes** or **variables** data. Variables data are usually continuous measurements, such as length, voltage, or viscosity. Attributes data, on the other hand, are usually discrete data, often taking the form of counts. Such as the number of loan applications that could not be properly processed because of missing required information, or the number of emergency room arrivals that have to wait more than 30 minutes to receive medical attention. We will describe statistical-based quality engineering tools for dealing with both types of data.

Quality characteristics are often evaluated relative to **specifications**. For a manufactured product, the specifications are the desired measurements for the quality characteristics of the components and subassemblies that make up the product, as well as the desired values for the quality characteristics in the final product. For example, the diameter of a shaft used in an automobile transmission cannot be too large or it will not fit into the mating bearing, nor can it be too small, resulting in a loose fit, causing vibration, wear, and early failure of the assembly. In the service industries, specifications are typically in terms of the maximum amount of time to process an order or to provide a particular service.

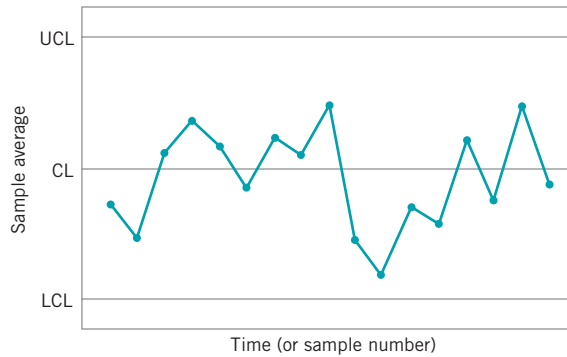
1.3 Statistical Methods for Quality Control and Improvement

This textbook concentrates on statistical and engineering technology useful in quality improvement. Specifically, we focus on three major areas: **statistical process control**, **design of experiments**, and (to a lesser extent) **acceptance sampling**. In addition to these techniques, a number of other statistical and analytical tools are useful in analyzing quality problems and improving the performance of processes. The role of some of these tools is illustrated in Fig. 1.3, which presents a **process** as a system with a set of inputs and an output. In the case of a manufacturing process, the controllable input factors $x_1, x_2, \dots, x_p$ are process variable such as temperatures, pressures, feed rates, and other process variables. The inputs $z_1, z_2, \dots, z_q$ are uncontrollable (or difficult to control) inputs, such as environmental factors or properties of raw materials provided by an external supplier. The production process transforms the input raw materials, component parts, and subassemblies into a finished product that has several quality characteristics. The output variable y is a quality characteristic, that is, a measure of process and product quality. This model can also be used to represent **non-manufacturing or service processes**. For example, consider a process in a financial institution that processes automobile loan applications. The inputs are the loan applications, which contain information about the customer and his/her credit history, the type of car to be purchased, its price, and the loan amount. The controllable factors are the type of training that the loan officer receives, the specific rules and policies that the bank imposed on these loans, and the number of people working as loan officers at each time period. The uncontrollable factors include prevailing interest rates, the amount of capital available for these types of loans in each time period, and the number of loan applications that require processing each period. The output quality characteristics include whether or not the loan is funded, the number of funded loans that are actually accepted by the applicant, and the cycle time; that is, the length of time that the customer waits until a decision on his/her loan application is made. In service systems, cycle time is often a very important CTQ.

A **control chart** is one of the primary techniques of **statistical process control (SPC)**. A typical control chart is shown in Fig. 1.4. This chart plots the averages of measurements of a quality characteristic in samples taken from the process versus time (or the sample number). The chart has a center line (CL) and upper and lower control limits (UCL and LCL in Fig. 1.4). The center line represents where this process characteristic should fall if there are no unusual sources of variability present. The control limits are determined from some simple statistical considerations that we will discuss in Chapters 4, 5, and 6. Classically, control



■ **FIGURE 1.3** Production process inputs and outputs.



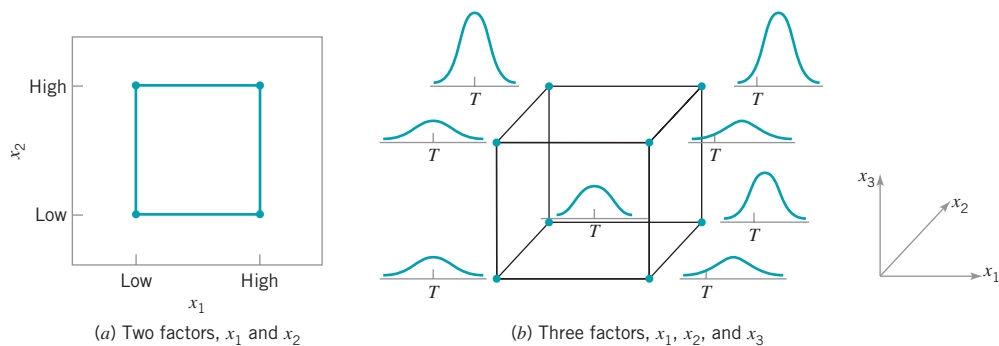
■ FIGURE 1.4 A typical control chart.

charts are applied to the output variable(s) in a system such as in Fig. 1.4. However, in some cases they can be usefully applied to the inputs as well.

The control chart is a very useful **process monitoring technique**; when unusual sources of variability are present, sample averages will plot outside the control limits. This is a signal that some investigation of the process should be made and corrective action to remove these unusual sources of variability taken. Systematic use of a control chart is an excellent way to reduce variability.

A **designed experiment** is extremely helpful in discovering the key variables influencing the quality characteristics of interest in the process. A designed experiment is an approach to systematically varying the controllable input factors in the process and determining the effect these factors have on the output product parameters. Statistically designed experiments are invaluable in reducing the variability in the quality characteristics and in determining the levels of the controllable variables that optimize process performance. Often significant breakthroughs in process performance and product quality also result from using designed experiments.

One major type of designed experiment is the **factorial design**, in which factors are varied together in such a way that all possible combinations of factor levels are tested. Figure 1.5 shows two possible factorial designs for the process in Fig. 1.3, for the cases of $p = 2$ and $p = 3$ controllable factors. In Fig. 1.5a, the factors have two levels, low and high, and the four possible test combinations in this factorial experiment form the corners of a square. In Fig. 1.5b, there are three factors each at two levels, giving an experiment with eight test combinations arranged at the corners of a cube. The distributions at the corners of the cube represent the process performance at each combination of the controllable factors x_1 , x_2 , and x_3 . It is clear that some combinations of factor levels produce better results than others. For example, increasing x_1 from low to high increases the average level of the process output and could



■ FIGURE 1.5 Factorial designs for the process in Fig. 1.3.

shift it off the target value (T). Furthermore, process variability seems to be substantially reduced when we operate the process along the back edge of the cube, where x_2 and x_3 are at their high levels.

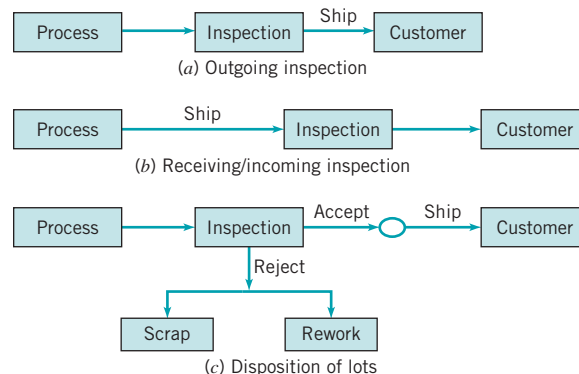
Designed experiments are a major **off-line** quality-control tool, because they are often used during development activities and the early stages of manufacturing, rather than as a routine **on-line** or **in-process** procedure. They play a crucial role in reducing variability.

Once we have identified a list of important variables that affect the process output, it is usually necessary to model the relationship between the influential input variables and the output quality characteristics. Statistical techniques useful in constructing such models include regression analysis and time series analysis. Detailed discussions of designed experiments, regression analysis, and time series modeling are in Montgomery (2005), Montgomery, Peck, and Vining (2006), and Box, Jenkins, and Reinsel (1994).

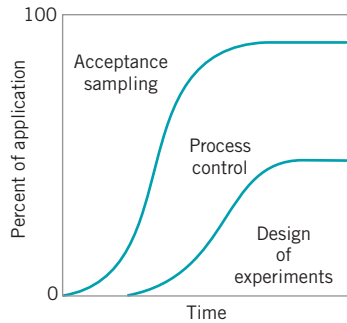
When the important variables have been identified and the nature of the relationship between the important variables and the process output has been quantified, then an on-line statistical process-control technique for monitoring and surveillance of the process can be employed with considerable effectiveness. Techniques such as control charts can be used to monitor the process output and detect when changes in the inputs are required to bring the process back to an in-control state. The models that relate the influential inputs to process outputs help determine the nature and magnitude of the adjustments required. In many processes, once the dynamic nature of the relationships between the inputs and the outputs are understood, it may be possible to routinely adjust the process so that future values of the product characteristics will be approximately on target. This routine adjustment is often called **engineering control, automatic control, or feedback control**. We will briefly discuss these types of process control schemes in Chapter 11 and illustrate how statistical process control (or **SPC**) methods can be successfully integrated into a manufacturing system in which engineering control is in use.

The third area of quality control and improvement that we discuss is **acceptance sampling**. This is closely connected with inspection and testing of product, which is one of the earliest aspects of quality control, dating back to long before statistical methodology was developed for quality improvement. Inspection can occur at many points in a process. Acceptance sampling, defined as the inspection and classification of a sample of units selected at random from a larger batch or lot and the ultimate decision about disposition of the lot, usually occurs at two points: incoming raw materials or components, or final production.

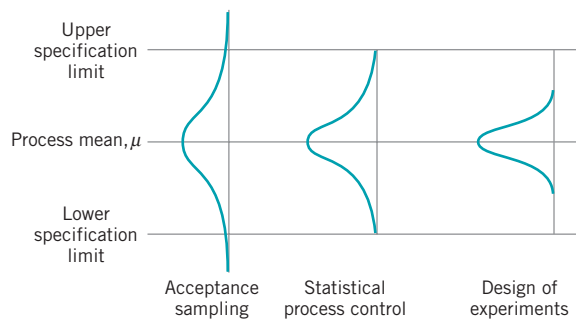
Several different variations of acceptance sampling are shown in Fig. 1.6. In Fig. 1.6a, the inspection operation is performed immediately following production, before the product is shipped to the customer. This is usually called **outgoing inspection**. Figure 1.6b illustrates **incoming inspection**; that is, a situation in which lots of batches of product are sampled as they are received from the supplier. Various lot-dispositioning decisions are illustrated in Fig. 1.6c.



■ **FIGURE 1.6** Variations of acceptance sampling.



■ **FIGURE 1.7** Phase diagram of the use of quality-engineering methods.



■ **FIGURE 1.8** Application of quality-engineering techniques and the systematic reduction of process variability.

Sampled lots may either be accepted or rejected. Items in a rejected lot are typically either scrapped or recycled, or they may be reworked or replaced with good units. This latter case is often called **rectifying inspection**.

Modern quality assurance systems usually place less emphasis on acceptance sampling and attempt to make statistical process control and designed experiments the focus of their efforts. Acceptance sampling tends to reinforce the conformance-to-specification view of quality and does not have any feedback into either the production process or engineering design or development that would necessarily lead to quality improvement.

Figure 1.7 shows the typical evolution in the use of these techniques in most organizations. At the lowest level of maturity, management may be completely unaware of quality issues, and there is likely to be no effective organized quality improvement effort. Frequently there will be some modest applications of acceptance-sampling and inspection methods, usually for incoming parts and materials. The first activity as maturity increases is to intensify the use of sampling inspection. The use of sampling will increase until it is realized that quality cannot be inspected or tested into the product.

At that point, the organization usually begins to focus on process improvement. Statistical process control and experimental design potentially have major impacts on manufacturing, product design activities, and process development. The systematic introduction of these methods usually marks the start of substantial quality, cost, and productivity improvements in the organization. At the highest levels of maturity, companies use designed experiments and statistical process control methods intensively and make relatively modest use of acceptance sampling.

The primary **objective** of quality engineering efforts is the **systematic reduction of variability** in the key quality characteristics of the product. Figure 1.8 shows how this happens over time. In the early stages, when acceptance sampling is the major technique in use, process “fallout,” or units that do not conform to the specifications, constitute a high percentage of the process output. The introduction of statistical process control will stabilize the process and reduce the variability. However, it is not satisfactory just to meet requirements—further reduction of variability usually leads to better product performance and enhanced competitive position, as was vividly demonstrated in the automobile transmission example discussed earlier. Statistically designed experiments can be employed in conjunction with statistical process monitoring and control to minimize process variability in nearly all industrial settings.

1.4 Management Aspects of Quality Improvement

Statistical techniques, including SPC and designed experiments, along with other problem-solving tools are the technical basis for quality control and improvement. However, to be used most effectively, these techniques must be implemented within and be part of a management

system that is focused on quality improvement. The management system of an organization must be organized to properly direct the overall quality improvement philosophy and ensure its deployment in all aspects of the business. The effective management of quality involves successful execution of three activities: quality planning, quality assurance, and quality control and improvement.

Quality planning is a strategic activity, and it is just as vital to an organization's long-term business success as the product development plan, the financial plan, the marketing plan, and plans for the utilization of human resources. Without a strategic quality plan, an enormous amount of time, money, and effort will be wasted by the organization dealing with faulty designs, manufacturing defects, field failures, and customer complaints. Quality planning involves identifying customers, both external and those that operate internal to the business, and identifying their needs [this is sometimes called listening to the **voice of the customer (VOC)**]. Then products or services that meet or exceed customer expectations must be developed. The eight dimensions of quality discussed in Section 1.1.1 are an important part of this effort. The organization must then determine how these products and services will be realized. Planning for quality improvement on a specific, systematic basis is also a vital part of this process.

Quality assurance is the set of activities that ensures the quality levels of products and services are properly maintained and that supplier and customer quality issues are properly resolved. Documentation of the quality system is an important component. Quality system documentation involves four components: policy, procedures, work instructions and specifications, and records. Policy generally deals with what is to be done and why, while procedures focus on the methods and personnel that will implement policy. Work instructions and specifications are usually product-, department-, tool-, or machine-oriented. Records are a way of documenting the policies, procedures, and work instructions that have been followed. Records are also used to track specific units or batches of product, so that it can be determined exactly how they were produced. Records are often vital in providing data for dealing with customer complaints, corrective actions, and, if necessary, product recalls. Development, maintenance, and control of documentation are important quality assurance functions. One example of document control is ensuring that specifications and work instructions developed for operating personnel reflect the latest design and engineering changes.

Quality control and improvement involve the set of activities used to ensure that the products and services meet requirements and are improved on a continuous basis. Since variability is often a major source of poor quality, statistical techniques, including SPC and designed experiments, are the major tools of quality control and improvement. Quality improvement is often done on a project-by-project basis and involves teams led by personnel with specialized knowledge of statistical methods and experience in applying them. Projects should be selected so that they have significant business impact and are linked with the overall business goals for quality identified during the planning process. The techniques in this book are integral to successful quality control and improvement.

The next section provides a brief overview of some of the key elements of quality management. We discuss some of the important quality philosophies; quality systems and standards; the link between quality and productivity and quality and cost; economic and legal implications of quality; and some aspects of implementation. The three aspects of quality planning, quality assurance, and quality control and improvement will be woven into the discussion.

1.4.1 Quality Philosophy and Management Strategies

Many people have contributed to the statistical methodology of quality improvement. However, in terms of implementation and management philosophy, three individuals emerge as the leaders: W. E. Deming, J. M. Juran, and A. V. Feigenbaum. We now briefly discuss the approaches and philosophy of those leaders in quality management.

of statistical tools need to be widespread. However, the organizational aspects of Feigenbaum's work are important, as quality improvement does not usually spring forth as a "grass roots" activity; it requires a lot of management commitment to make it work.

The brief descriptions of the philosophies of Deming, Juran, and Feigenbaum have highlighted both the common aspects and differences of their viewpoints. In this author's opinion, there are more similarities than differences among them, and the similarities are what is important. All three of these pioneers stress the importance of quality as an essential competitive weapon, the important role that management must play in implementing quality improvement, and the importance of statistical methods and techniques in the "quality transformation" of an organization.

Total Quality Management. Total quality management (TQM) is a strategy for implementing and managing quality improvement activities on an organizationwide basis. TQM began in the early 1980s, with the philosophies of Deming and Juran as the focal point. It evolved into a broader spectrum of concepts and ideas, involving participative organizations and work culture, customer focus, supplier quality improvement, integration of the quality system with business goals, and many other activities to focus all elements of the organization around the quality improvement goal. Typically, organizations that have implemented a TQM approach to quality improvement have quality councils or high-level teams that deal with strategic quality initiatives, workforce-level teams that focus on routine production or business activities, and cross-functional teams that address specific quality improvement issues.

TQM has only had moderate success for a variety of reasons, but frequently because there is insufficient effort devoted to widespread utilization of the technical tools of variability reduction. Many organizations saw the mission of TQM as one of training. Consequently, many TQM efforts engaged in widespread training of the workforce in the philosophy of quality improvement and a few basic methods. This training was usually placed in the hands of human resources departments, and much of it was ineffective. The trainers often had no real idea about *what* methods should be taught, and success was usually measured by the percentage of the workforce that had been "trained," not by whether any measurable impact on business results had been achieved. Some general reasons for the lack of conspicuous success of TQM include (1) lack of topdown, high-level **management commitment and involvement**; (2) inadequate use of statistical methods and insufficient recognition of variability reduction as a prime objective; (3) general as opposed to specific business-results-oriented objectives; and (4) too much emphasis on widespread **training** as opposed to focused technical **education**.

Another reason for the erratic success of TQM is that many managers and executives have regarded it as just another "program" to improve quality. During the 1950s and 1960s, programs such as **Zero Defects** and **Value Engineering** abounded, but they had little real impact on quality and productivity improvement. During the heyday of TQM in the 1980s, another popular program was the **Quality is Free** initiative, in which management worked on identifying the cost of quality (or the cost of nonquality, as the Quality is Free devotees so cleverly put it). Indeed, identification of quality costs can be very useful (we discuss quality costs in Section 1.4.3), but the Quality is Free practitioners often had no idea about what to do to actually improve many types of complex industrial processes. In fact, the leaders of this initiative had no knowledge about statistical methodology and completely failed to understand its role in quality improvement. When TQM is wrapped around an ineffective program such as this, disaster is often the result.

Quality Systems and Standards. The International Standards Organization (founded in 1946 in Geneva, Switzerland), known as ISO, has developed a series of standards for quality systems. The first standards were issued in 1987. The current version of the standard is known as the ISO 9000 series. It is a generic standard, broadly applicable to any type

of organization, and it is often used to demonstrate a supplier's ability to control its processes. The three standards of ISO 9000 are:

ISO 9000:2000 Quality Management System—Fundamentals and Vocabulary

ISO 9001:2000 Quality Management System—Requirements

ISO 9004:2000 Quality Management System—Guidelines for Performance Improvement

ISO 9000 is also an American National Standards Institute and an ASQ standard.

The ISO 9001:2000 standard has eight clauses: (1) Scope, (2) Normative References, (3) Definitions, (4) Quality Management Systems, (5) Management Responsibility, (6) Resource Management, (7) Product (or Service) Realization, and (8) Measurement, Analysis, and Improvement. Clauses 4 through 8 are the most important, and their key components and requirements are shown in Table 1.3. To become certified under the ISO standard, a company must select a **registrar** and prepare for a **certification audit** by this registrar. There is no single independent authority that licenses, regulates, monitors, or qualifies registrars. As we will discuss later, this is a serious problem with the ISO system. Preparing for the certification audit involves many activities, including (usually) an initial or phase I audit that checks the present quality management system against the standard. This is usually followed by establishing teams to ensure that all components of the key clause are developed and implemented, training of personnel, developing applicable documentation, and developing and installing all new components of the quality system that may be required. Then the certification audit takes place. If the company is certified, then periodic **surveillance audits** by the registrar continue, usually on an annual (or perhaps six-month) schedule.

Many organizations have required their suppliers to become certified under ISO 9000, or one of the standards that are more industry-specific. Examples of these industry-specific quality system standards are AS 9100 for the aerospace industry; ISO/TS 16949 and QS 9000 for the automotive industry; and TL 9000 for the telecommunications industry. Many components of these standards are very similar to those of ISO 9000.

Much of the focus of ISO 9000 (and of the industry-specific standards) is on formal documentation of the quality system; that is, on **quality assurance** activities. Organizations usually must make extensive efforts to bring their documentation into line with the requirements of the standards; this is the Achilles' heel of ISO 9000 and other related or derivative standards. There is far too much effort devoted to documentation, paperwork, and bookkeeping and not nearly enough to actually reducing variability and improving processes and products. Furthermore, many of the third-party registrars, auditors, and consultants that work in this area are not sufficiently educated or experienced enough in the **technical** tools required for **quality improvement** or how these tools should be deployed. They are all too often unaware of what constitutes modern engineering and statistical practice, and usually are familiar with only the most elementary techniques. Therefore, they concentrate largely on the documentation, record keeping, and paperwork aspects of certification.

There is also evidence that ISO certification or certification under one of the other industry-specific standards does little to prevent poor quality products from being designed, manufactured, and delivered to the customer. For example, in 1999–2000, there were numerous incidents of rollover accidents involving Ford Explorer vehicles equipped with Bridgestone/Firestone tires. There were nearly 300 deaths in the United States alone attributed to these accidents, which led to a recall by Bridgestone/Firestone of approximately 6.5 million tires. Apparently, many of the tires involved in these incidents were manufactured at the Bridgestone/Firestone plant in Decatur, Illinois. In an article on this story in *Time* magazine (September 18, 2000), there was a photograph (p. 38) of the sign at the entrance of the Decatur plant which stated that the plant was “QS 9000 Certified” and “ISO 14001 Certified” (ISO 14001 is an environmental standard).

■ **TABLE 1.3**
ISO 9001:2000 Requirements

4.0	Quality Management System
4.1	General Requirements The organization shall establish, document, implement, and maintain a quality management system and continually improve its effectiveness in accordance with the requirements of the international standard.
4.2	Documentation Requirements Quality management system documentation will include a quality policy and quality objectives; a quality manual; documented procedures; documents to ensure effective planning, operation, and control of processes; and records required by the international standard.
5.0	Management System
5.1	Management Commitment a. Communication of meeting customer, statutory, and regulatory requirements b. Establishing a quality policy c. Establishing quality objectives d. Conducting management reviews e. Ensuring that resources are available
5.2	Top management shall ensure that customer requirements are determined and are met with the aim of enhancing customer satisfaction.
5.3	Management shall establish a quality policy.
5.4	Management shall ensure that quality objectives shall be established. Management shall ensure that planning occurs for the quality management system.
5.5	Management shall ensure that responsibilities and authorities are defined and communicated.
5.6	Management shall review the quality management system at regular intervals.
6.0	Resource Management
6.1	The organization shall determine and provide needed resources.
6.2	Workers will be provided necessary education, training, skills, and experience.
6.3	The organization shall determine, provide, and maintain the infrastructure needed to achieve conformity to product requirements.
6.4	The organization shall determine and manage the work environment needed to achieve conformity to product requirements.
7.0	Product or Service Realization
7.1	The organization shall plan and develop processes needed for product or service realization.
7.2	The organization shall determine requirements as specified by customers.
7.3	The organization shall plan and control the design and development for its products or services.
7.4	The organization shall ensure that purchased material or product conforms to specified purchase requirements.
7.5	The organization shall plan and carry out production and service under controlled conditions.
7.6	The organization shall determine the monitoring and measurements to be undertaken and the monitoring and measuring devices needed to provide evidence of conformity of products or services to determined requirements.
8.0	Measurement, Analysis, and Improvement
8.1	The organization shall plan and implement the monitoring, measurement, analysis, and improvement process for continual improvement and conformity to requirements.
8.2	The organization shall monitor information relating to customer perceptions.
8.3	The organization shall ensure that product that does not conform to requirements is identified and controlled to prevent its unintended use or delivery.
8.4	The organization shall determine, collect, and analyze data to demonstrate the suitability and effectiveness of the quality management system, including a. Customer satisfaction b. Conformance data c. Trend data d. Supplier data
8.5	The organization shall continually improve the effectiveness of the quality management system.

Adapted from the ISO 9001:2000 Standard, International Standards Organization, Geneva, Switzerland, 2003.

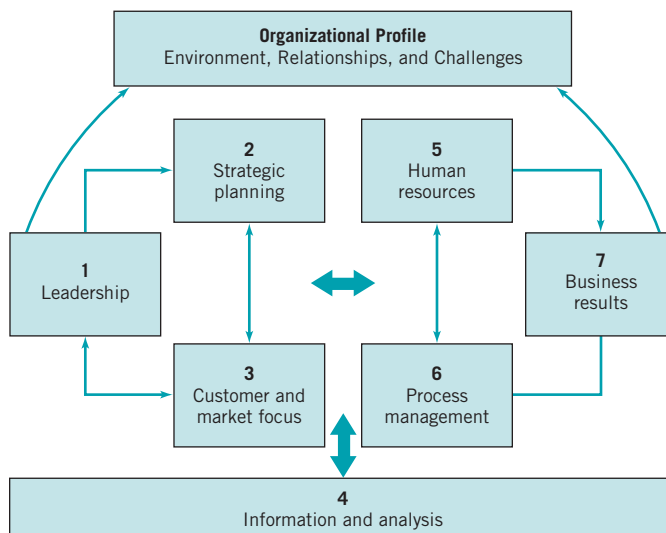
Although the assignable causes underlying these incidents have not been fully discovered, there are clear indicators that despite quality systems certification, Bridgestone/Firestone experienced significant quality problems. ISO certification alone is no guarantee that good quality products are being designed, manufactured, and delivered to the customer. Relying on ISO certification is a strategic management mistake.

It has been estimated that ISO certification activities are approximately a *\$40 billion annual business*, worldwide. Much of this money flows to the registrars, auditors, and consultants. This amount does not include all of the internal costs incurred by organizations to achieve registration, such as the thousands of hours of engineering and management effort, travel, internal training, and internal auditing. It is not clear whether any significant fraction of this expenditure has made its way to the bottom line of the registered organizations. Furthermore, there is no assurance that certification has any real impact on quality (as in the Bridgestone/Firestone tire incidents). Many quality engineering authorities believe that ISO certification is largely a waste of effort. Often, organizations would be far better off to “just say no to ISO” and spend a small fraction of that \$40 billion on their quality systems and another larger fraction on meaningful variability reduction efforts, develop their own internal (or perhaps industry-based) quality standards, rigorously enforce them, and pocket the difference.

The Malcolm Baldrige National Quality Award. The Malcolm Baldrige National Quality Award (MBNQA) was created by the U.S. Congress in 1987. It is given annually to recognize U.S. organizations for performance excellence. Awards are given to organizations in five categories: manufacturing, service, small business, health care, and education. Three awards may be given each year in each category. Many organizations compete for the awards, and many companies use the performance excellence criteria for self-assessment. The award is administered by NIST (the National Bureau of Standards and Technology).

The performance excellence criteria and their interrelationships are shown in Fig. 1.10. The point values for these criteria in the MBNQA are shown in Table 1.4. The criteria are directed towards results, where results are a composite of customer satisfaction and retention, market share and new market development, product/service quality, productivity and operational effectiveness, human resources development, supplier performance, and public/corporate citizenship. The criteria are nonprescriptive. That is, the focus is on results, not the use of specific procedures or tools.

The MBNQA process is shown in Fig. 1.11. An applicant sends the completed application to NIST. This application is then subjected to a first-round review by a team of Baldrige



■ **FIGURE 1.10** The structure of the MBNQA performance excellence criteria. (Source: Foundation for the Malcolm Baldrige National Quality Award, 2002 Criteria for Performance Excellence.)

Note that the total yield from this process, after reworking, is 90 good parts per day.

An engineering study of this process reveals that excessive process variability is responsible for the extremely high fallout. A new statistical process-control procedure is implemented that reduces variability, and consequently the process fallout decreases from 25% to 5%. Of the 5% fallout produced, about 60% can be reworked, and 40% are scrapped. After the process-control program is implemented, the manufacturing cost per good part produced is

$$\text{Cost/good part} = \frac{\$20(100) + \$4(3)}{98} = \$20.53$$

Note that the installation of statistical process control and the reduction of variability that follows result in a 10.3% reduction in manufacturing costs. Furthermore, productivity is up by almost 10%; 98 good parts are produced each day as opposed to 90 good parts previously. This amounts to an increase in production capacity of almost 10%, without any additional investment in equipment, workforce, or overhead. Efforts to improve this process by other methods (such as Just-in-Time, lean manufacturing, etc.) are likely to be completely ineffective until the basic problem of excessive variability is solved.

1.4.3 Quality Costs

Financial controls are an important part of business management. These financial controls involve a comparison of actual and budgeted costs, along with analysis and action on the differences between actual and budget. It is customary to apply these financial controls on a department or functional level. For many years, there was no direct effort to measure or account for the costs of the quality function. However, many organizations now formally evaluate the cost associated with quality. There are several reasons why the cost of quality should be explicitly considered in an organization. These include the following:

1. The increase in the cost of quality because of the increase in the complexity of manufactured products associated with advances in technology
2. Increasing awareness of life-cycle costs, including maintenance, spare parts, and the cost of field failures
3. Quality engineers and managers can most effectively communicate quality issues in a way that management understands.

As a result, quality costs have emerged as a financial control tool for management and as an aid in identifying opportunities for reducing quality costs.

Generally speaking, quality costs are those categories of costs that are associated with producing, identifying, avoiding, or repairing products that do not meet requirements. Many manufacturing and service organizations use four categories of quality costs: prevention costs, appraisal costs, internal failure costs, and external failure costs. These cost categories are shown in Table 1.5. We now discuss these categories in more detail.

Prevention Costs. Prevention costs are those costs associated with efforts in design and manufacturing that are directed toward the prevention of nonconformance. Broadly speaking, prevention costs are all costs incurred in an effort to “make it right the first time.” The important subcategories of prevention costs follow.

Quality planning and engineering. Costs associated with the creation of the overall quality plan, the inspection plan, the reliability plan, the data system, and all specialized plans and activities of the quality-assurance function; the preparation of manuals and procedures used to communicate the quality plan; and the costs of auditing the system.

New products review. Costs of the preparation of bid proposals, the evaluation of new designs from a quality viewpoint, the preparation of tests and experimental programs to

■ TABLE 1.5

Quality Costs

Prevention Costs	Internal Failure Costs
Quality planning and engineering	Scrap
New products review	Rework
Product/process design	Retest
Process control	Failure analysis
Burn-in	Downtime
Training	Yield losses
Quality data acquisition and analysis	Downgrading (off-specing)
Appraisal Costs	External Failure Costs
Inspection and test of incoming material	Complaint adjustment
Product inspection and test	Returned product/material
Materials and services consumed	Warranty charges
Maintaining accuracy of test equipment	Liability costs
	Indirect costs

evaluate the performance of new products, and other quality activities during the development and preproduction stages of new products or designs.

Product/process design. Costs incurred during the design of the product or the selection of the production processes that are intended to improve the overall quality of the product. For example, an organization may decide to make a particular circuit component redundant because this will increase the reliability of the product by increasing the mean time between failures. Alternatively, it may decide to manufacture a component using process A rather than process B, because process A is capable of producing the product at tighter tolerances, which will result in fewer assembly and manufacturing problems. This may include a vendor's process, so the cost of dealing with other than the lowest bidder may also be a prevention cost.

Process control. The cost of process-control techniques, such as control charts, that monitor the manufacturing process in an effort to reduce variation and build quality into the product.

Burn-in. The cost of preshipment operation of the product to prevent early-life failures in the field.

Training. The cost of developing, preparing, implementing, operating, and maintaining formal training programs for quality.

Quality data acquisition and analysis. The cost of running the quality data system to acquire data on product and process performance; also the cost of analyzing these data to identify problems. It includes the work of summarizing and publishing quality information for management.

Appraisal Costs. Appraisal costs are those costs associated with measuring, evaluating, or auditing products, components, and purchased materials to ensure conformance to the standards that have been imposed. These costs are incurred to determine the condition of the product from a quality viewpoint and ensure that it conforms to specifications. The major subcategories follow.

Inspection and test of incoming material. Costs associated with the inspection and testing of all material. This subcategory includes receiving inspection and test; inspection, test, and evaluation at the vendor's facility; and a periodic audit of the quality-assurance system. This could also include intraplant vendors.

Product inspection and test. The cost of checking the conformance of the product throughout its various stages of manufacturing, including final acceptance testing, packing and shipping checks, and any test done at the customer's facilities prior to turning the product over to the customer. This also includes life testing, environmental testing, and reliability testing.

Materials and services consumed. The cost of material and products consumed in a destructive test or devalued by reliability tests.

Maintaining accuracy of test equipment. The cost of operating a system that keeps the measuring instruments and equipment in calibration.

Internal Failure Costs. Internal failure costs are incurred when products, components, materials, and services fail to meet quality requirements, and this failure is discovered prior to delivery of the product to the customer. These costs would disappear if there were no defects in the product. The major subcategories of internal failure costs follow.

Scrap. The net loss of labor, material, and overhead resulting from defective product that cannot economically be repaired or used.

Rework. The cost of correcting nonconforming units so that they meet specifications. In some manufacturing operations rework costs include additional operations or steps in the manufacturing process that are created to solve either chronic defects or sporadic defects.

Retest. The cost of reinspection and retesting of products that have undergone rework or other modifications.

Failure analysis. The cost incurred to determine the causes of product failures.

Downtime. The cost of idle production facilities that results from nonconformance to requirements. The production line may be down because of nonconforming raw materials supplied by a supplier, which went undiscovered in receiving inspection.

Yield losses. The cost of process yields that are lower than might be attainable by improved controls (for example, soft-drink containers that are overfilled because of excessive variability in the filling equipment).

Downgrading/off-specing. The price differential between the normal selling price and any selling price that might be obtained for a product that does not meet the customer's requirements. Downgrading is a common practice in the textile, apparel goods, and electronics industries. The problem with downgrading is that products sold do not recover the full contribution margin to profit and overhead as do products that conform to the usual specifications.

External Failure Costs. External failure costs occur when the product does not perform satisfactorily after it is delivered to the customer. These costs would also disappear if every unit of product conformed to requirements. Subcategories of external failure costs follow.

Complaint adjustment. All costs of investigation and adjustment of justified complaints attributable to the nonconforming product.

Returned product/material. All costs associated with receipt, handling, and replacement of the nonconforming product or material that is returned from the field.

Warranty charges. All costs involved in service to customers under warranty contracts.

Liability costs. Costs or awards incurred from product liability litigation.

Indirect costs. In addition to direct operating costs of external failures, there are a significant number of indirect costs. These are incurred because of customer dissatisfaction with the level of quality of the delivered product. Indirect costs may reflect the customer's attitude toward

the company. They include the costs of loss of business reputation, loss of future business, and loss of market share that inevitably results from delivering products and services that do not conform to the customer's expectations regarding fitness for use.

The Analysis and Use of Quality Costs. How large are quality costs? The answer, of course, depends on the type of organization and the success of their quality improvement effort. In some organizations quality costs are 4% or 5% of sales, whereas in others they can be as high as 35% or 40% of sales. Obviously, the cost of quality will be very different for a high-technology computer manufacturer than for a typical service industry, such as a department store or hotel chain. In most organizations, however, quality costs are higher than necessary, and management should make continuing efforts to appraise, analyze, and reduce these costs.

The usefulness of quality costs stems from the **leverage effect**; that is, dollars invested in prevention and appraisal have a payoff in reducing dollars incurred in internal and external failures that exceeds the original investment. For example, a dollar invested in prevention may return \$10 or \$100 (or more) in savings from reduced internal and external failures.

Quality-cost analyses have as their principal objective cost reduction through identification of improvement opportunities. This is often done with a **Pareto analysis**. The Pareto analysis consists of identifying quality costs by category, or by product, or by type of defect or nonconformity. For example, inspection of the quality-cost information in Table 1.6 concerning defects or nonconformities in the assembly of electronic components onto printed circuit boards reveals that insufficient solder is the highest quality cost incurred in this operation. Insufficient solder accounts for 42% of the total defects in this particular type of board and for almost 52% of the total scrap and rework costs. If the wave solder process can be improved, then there will be dramatic reductions in the cost of quality.

How much reduction in quality costs is possible? Although the cost of quality in many organizations can be significantly reduced, it is unrealistic to expect it can be reduced to zero. Before that level of performance is reached, the incremental costs of prevention and appraisal will rise more rapidly than the resulting cost reductions. However, paying attention to quality costs in conjunction with a focused effort on variability reduction has the capability of reducing quality costs by 50% or 60% provided that no organized effort has previously existed. This cost reduction also follows the Pareto principle; that is, most of the cost reductions will come from attacking the few problems that are responsible for the majority of quality costs.

In analyzing quality costs and in formulating plans for reducing the cost of quality, it is important to note the role of prevention and appraisal. Many organizations devote far too much effort to appraisal and not enough to prevention. This is an easy mistake for an organization to make, because appraisal costs are often budget line items in manufacturing. On the

■ **TABLE 1.6**
Monthly Quality-Costs Information for Assembly of Printed Circuit Boards

Type of Defect	Percent of Total Defects	Scrap and Rework Costs
Insufficient solder	42%	\$37,500.00 (52%)
Misaligned components	21	12,000.00
Defective components	15	8,000.00
Missing components	10	5,100.00
Cold solder joints	7	5,000.00
All other causes	5	4,600.00
Totals	100%	\$72,200.00

other hand, prevention costs may not be routinely budgeted items. It is not unusual to find in the early stages of a quality-cost program that appraisal costs are eight or ten times the magnitude of prevention costs. This is probably an unreasonable ratio, as dollars spent in prevention have a much greater payback than do dollars spent in appraisal.

Generating the quality-cost figures is not always easy, because most quality-cost categories are not a direct component in the accounting records of the organization. Consequently, it may be difficult to obtain extremely accurate information on the costs incurred with respect to the various categories. The organization's accounting system can provide information on those quality-cost categories that coincide with the usual business accounts, such as, for example, product testing and evaluation. In addition, many companies will have detailed information on various categories of failure cost. The information for cost categories for which exact accounting information is not available should be generated by using estimates, or, in some cases, by creating special monitoring and surveillance procedures to accumulate those costs over the study period.

The reporting of quality costs is usually done on a basis that permits straightforward evaluation by management. Managers want quality costs expressed in an index that compares quality cost with the opportunity for quality cost. Consequently, the usual method of reporting quality costs is in the form of a ratio, where the numerator is quality-cost dollars and the denominator is some measure of activity, such as (1) hours of direct production labor, (2) dollars of direct production labor, (3) dollars of processing costs, (4) dollars of manufacturing cost, (5) dollars of sales, or (6) units of product.

Upper management may want a standard against which to compare the current quality-cost figures. It is difficult to obtain absolute standards and almost as difficult to obtain quality-cost levels of other companies in the same industry. Therefore, the usual approach is to compare current performance with past performance so that, in effect, quality-cost programs report variances from past performance. These trend analyses are primarily a device for detecting departures from standard and for bringing them to the attention of the appropriate managers. They are not necessarily in and of themselves a device for ensuring quality improvements.

This brings us to an interesting observation: Some quality-cost collection and analysis efforts fail. That is, a number of companies have started quality-cost analysis activities, used them for some time, and then abandoned the programs as ineffective. There are several reasons why this occurs. Chief among these is failure to use quality-cost information as a mechanism for generating improvement opportunities. If we use quality cost information as a scorekeeping tool only, and do not make conscious efforts to identify problem areas and develop improved operating procedures and processes, then the programs will not be totally successful.

Another reason why quality-cost collection and analysis doesn't lead to useful results is that managers become preoccupied with perfection in the cost figures. Overemphasis in treating quality costs as part of the accounting systems rather than as a management control tool is a serious mistake. This approach greatly increases the amount of time required to develop the cost data, analyze them, and identify opportunities for quality improvements. As the time required to generate and analyze the data increases, management becomes more impatient and less convinced of the effectiveness of the activity. Any program that appears to management as going nowhere is likely to be abandoned.

A final reason for the failure of a quality-cost program is that management often underestimates the depth and extent of the commitment to prevention that must be made. The author has had numerous opportunities to examine quality cost data in many companies. In companies without effective quality improvement programs, the dollars allocated to prevention rarely exceed 1% to 2% of revenue. This must be increased to a threshold of about 5% to 6% of revenue, and these additional prevention dollars must be spent largely on the technical methods of quality improvement, and not on establishing programs such as TQM, Zero Defects, or other

similar activities. If management is persistent in this effort, then the cost of quality will decrease substantially. These cost savings will typically begin to occur in one to two years, although it could be longer in some companies.

1.4.4 Legal Aspects of Quality

Consumerism and product liability are important reasons why quality assurance is an important business strategy. Consumerism is in part due to the seemingly large number of failures in the field of consumer products and the perception that service quality is declining. Highly visible field failures often prompt the questions of whether today's products are as good as their predecessors and whether manufacturers are really interested in quality. The answer to both of these questions is yes. Manufacturers are always vitally concerned about field failures because of heavy external failure costs and the related threat to their competitive position. Consequently, most producers have made product improvements directed toward reducing field failures. For example, solid-state and integrated-circuit technology has greatly reduced the failure of electronic equipment that once depended on the electron tube. Virtually every product line of today is superior to that of yesterday.

Consumer dissatisfaction and the general feeling that today's products are inferior to their predecessors arise from other phenomena. One of these is the explosion in the number of products. For example, a 1% field-failure rate for a consumer appliance with a production volume of 50,000 units per year means 500 field failures. However, if the production rate is 500,000 units per year and the field-failure rate remains the same, then 5,000 units will fail in the field. This is equivalent, in the total number of dissatisfied customers, to a 10% failure rate at the lower production level. Increasing production volume increases the **liability exposure** of the manufacturer. Even in situations in which the failure rate declines, if the production volume increases more rapidly than the decrease in failure rate, the total number of customers who experience failures will still increase.

A second aspect of the problem is that consumer tolerance for minor defects and aesthetic problems has decreased considerably, so that blemishes, surface-finish defects, noises, and appearance problems that were once tolerated now attract attention and result in adverse consumer reaction. Finally, the competitiveness of the marketplace forces many manufacturers to introduce new designs before they are fully evaluated and tested in order to remain competitive. These "early releases" of unproved designs are a major reason for new product quality failures. Eventually, these design problems are corrected, but the high failure rate connected with new products often supports the belief that today's quality is inferior to that of yesterday.

Product liability is a major social, market, and economic force. The legal obligation of manufacturers and sellers to compensate for injury or damage caused by defective products is not a recent phenomenon. The concept of product liability has been in existence for many years, but its emphasis has changed recently. The first major product liability case occurred in 1916 and was tried before the New York Court of Appeals. The court held that an automobile manufacturer had a product liability obligation to a car buyer, even though the sales contract was between the buyer and a third party—namely, a car dealer. The direction of the law has always been that manufacturers or sellers are likely to incur a liability when they have been unreasonably careless or negligent in what they have designed, or produced, or how they have produced it. In recent years, the courts have placed a more stringent rule in effect called **strict liability**. Two principles are characteristic of strict liability. The first is a strong responsibility for both manufacturer and merchandiser, requiring immediate responsiveness to unsatisfactory quality through product service, repair, or replacement of defective product. This extends into the period of actual use by the consumer. By producing a product, the manufacturer and seller must accept responsibility for the ultimate use of that product—not only for its performance, but also for its environmental effects, the safety aspects of its use, and so forth.

The second principle involves advertising and promotion of the product. Under strict product liability all advertising statements must be supportable by valid company quality or certification data, comparable to that now maintained for product identification under regulations for such products as automobiles.

These two strict product liability principles result in strong pressure on manufacturers, distributors, and merchants to develop and maintain a high degree of factually based evidence concerning the performance and safety of their products. This evidence must cover not only the quality of the product as it is delivered to the consumer, but also its durability or reliability, its protection from possible side effects or environmental hazards, and its safety aspects in actual use. A strong quality-assurance program can help management in ensuring that this information will be available, if needed.

1.4.5 Implementing Quality Improvement

In the past few sections we have discussed the philosophy of quality improvement, the link between quality and productivity, and both economic and legal implications of quality. These are important aspects of the management of quality within an organization. There are certain other aspects of the overall management of quality that warrant some attention.

Management must recognize that quality is a multifaceted entity, incorporating the eight dimensions we discussed in Section 1.1.1. For convenient reference, Table 1.7 summarizes these quality dimensions.

A critical part of the **strategic management of quality** within any business is the recognition of these dimensions by management and the selection of dimensions along which the business will compete. It will be very difficult to compete against companies that can successfully accomplish this part of the strategy.

A good example is the Japanese dominance of the videocassette recorder (VCR) market. The Japanese did not invent the VCR; the first units for home use were designed and produced in Europe and North America. However, the early VCRs produced by these companies were very unreliable and frequently had high levels of manufacturing defects. When the Japanese entered the market, they elected to compete along the dimensions of reliability and conformance to standards (no defects). This strategy allowed them to quickly dominate the market. In subsequent years, they expanded the dimensions of quality to include added features, improved performance, easier serviceability, improved aesthetics, and so forth. They have used total quality as a competitive weapon to raise the entry barrier to this market so high that it is virtually impossible for a new competitor to enter.

Management must do this type of strategic thinking about quality. It is not necessary that the product be superior in all dimensions of quality, but management must **select and develop** the “niches” of quality along which the company can successfully compete. Typically, these dimensions will be those that the competition has forgotten or ignored. The American automobile industry has been severely impacted by foreign competitors who expertly practiced this strategy.

The critical role of **suppliers** in quality management must not be forgotten. In fact, supplier selection and **supply chain management** may be the most critical aspects of successful

■ **TABLE 1.7**
The Eight Dimensions of Quality from Section 1.1.1

1. Performance	5. Aesthetics
2. Reliability	6. Features
3. Durability	7. Perceived quality
4. Serviceability	8. Conformance to standards

quality management in industries such as automotive, aerospace, and electronics, where a very high percentage of the parts in the end item are manufactured by outside suppliers. Many companies have instituted formal supplier quality-improvement programs as part of their own **internal** quality-improvement efforts. Selection of suppliers based on **quality, schedule, and cost**, rather than on cost alone, is also a vital strategic management decision that can have a long-term significant impact on overall competitiveness.

It is also critical that management recognize that quality improvement must be a total, company-wide activity, and that every organizational unit *must* actively participate. Obtaining this participation is the responsibility of (and a significant challenge to) senior management. What is the role of the quality-assurance organization in this effect? The responsibility of quality assurance is to assist management in providing quality assurance for the companies' products. Specifically, the quality-assurance function is a technology warehouse that contains the skills and resources necessary to generate products of acceptable quality in the marketplace. Quality management also has the responsibility for evaluating and using quality-cost information for identifying improvement opportunities in the system, and for making these opportunities known to higher management. It is important to note, however, that the **quality function is not responsible for quality**. After all, the quality organization does not design, manufacture, distribute, or service the product. Thus, the responsibility for quality is distributed throughout the entire organization.

The philosophy of Deming, Juran, and Feigenbaum implies that responsibility for quality spans the entire organization. However, there is a danger that if we adopt the philosophy that "quality is everybody's job," then quality will become nobody's job. This is why quality planning and analysis are important. Because quality improvement activities are so broad, successful efforts require, as an initial step, top management commitment. This commitment involves emphasis on the importance of quality, identification of the respective quality responsibilities of the various organizational units, and explicit accountability for quality improvement of all managers and employees in the company.

Finally, strategic management of quality in an organization must involve all three components discussed earlier: **quality planning, quality assurance, and quality control and improvement**. Furthermore, *all* of the individuals in the organization must have an understanding of the basic tools of quality improvement. Central among these tools are the elementary statistical concepts that form the basis of process control and that are used for the analysis of process data. It is increasingly important that everyone in an organization, from top management to operating personnel, have an awareness of basic statistical methods and of how these methods are useful in manufacturing, engineering design, and development and in the general business environment. Certain individuals must have higher levels of skills; for example, those engineers and managers in the quality-assurance function would generally be experts in one or more areas of process control, reliability engineering, design of experiments, or engineering data analysis. However, the key point is the philosophy that statistical methodology is a language of communication about problems that enables management to mobilize resources rapidly and to efficiently develop solutions to such problems. Because six-sigma incorporates most of the elements for success that we have identified, it has proven to be a very effective framework for implementing quality improvement.

Important Terms and Concepts

Acceptance sampling
Appraisal costs
Critical-to-quality (CTQ)
Deming's 14 points

Designed experiments
Dimensions of quality
Fitness for use
Internal and external failure costs

5 *Methods and Philosophy of Statistical Process Control*

CHAPTER OUTLINE

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| 5.3.5 | Analysis of Patterns on Control Charts | | |
| 5.3.6 | Discussion of Sensitizing Rules for Control Charts | | |
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Supplemental Material for Chapter 5

- S5.1 A SIMPLE ALTERNATIVE TO RUNS RULES ON THE $\bar{x}$ CHART

The supplemental material is on the textbook Website www.wiley.com/college/montgomery.

CHAPTER OVERVIEW AND LEARNING OBJECTIVES

This chapter has three objectives. The first is to present the basic statistical control process (SPC) problem-solving tools, called the magnificent seven, and to illustrate how these tools form a cohesive, practical framework for quality improvement. These tools form an important basic approach to both reducing variability and monitoring the performance of a process, and are widely used in both the analyze and control steps of DMAIC. The second objective is to describe the statistical basis of the Shewhart control chart. The reader will see how decisions

about sample size, sampling interval, and placement of control limits affect the performance of a control chart. Other key concepts include the idea of rational subgroups, interpretation of control chart signals and patterns, and the average run length as a measure of control chart performance. The third objective is to discuss and illustrate some practical issues in the implementation of SPC.

After careful study of this chapter you should be able to do the following:

1. Understand chance and assignable causes of variability in a process
2. Explain the statistical basis of the Shewhart control chart, including choice of sample size, control limits, and sampling interval
3. Explain the rational subgroup concept
4. Understand the basic tools of SPC: the histogram or stem-and-leaf plot, the check sheet, the Pareto chart, the cause-and-effect diagram, the defect concentration diagram, the scatter diagram, and the control chart
5. Explain phase I and phase II use of control charts
6. Explain how average run length is used as a performance measure for a control chart
7. Explain how sensitizing rules and pattern recognition are used in conjunction with control charts

5.1 Introduction

If a product is to meet or exceed customer expectations, generally it should be produced by a process that is stable or repeatable. More precisely, the process must be capable of operating with little variability around the target or nominal dimensions of the product's quality characteristics. **Statistical process control (SPC)** is a powerful collection of problem-solving tools useful in achieving process stability and improving capability through the reduction of variability.

SPC is one of the greatest technological developments of the twentieth century because it is based on sound underlying principles, is easy to use, has significant impact, and can be applied to any process. Its seven major tools are

1. Histogram or stem-and-leaf plot
2. Check sheet
3. Pareto chart
4. Cause-and-effect diagram
5. Defect concentration diagram
6. Scatter diagram
7. Control chart

Although these tools, often called “**the magnificent seven**,” are an important part of SPC, they comprise only its technical aspects. The proper deployment of SPC helps create an environment in which all individuals in an organization seek continuous improvement in quality and productivity. This environment is best developed when management becomes involved in the process. Once this environment is established, routine application of the magnificent seven becomes part of the usual manner of doing business, and the organization is well on its way to achieving its quality improvement objectives.

Of the seven tools, the Shewhart control chart is probably the most technically sophisticated. It was developed in the 1920s by Walter A. Shewhart of the Bell Telephone

Laboratories. To understand the statistical concepts that form the basis of SPC, we must first describe Shewhart's theory of variability.

5.2 Chance and Assignable Causes of Quality Variation

In any production process, regardless of how well designed or carefully maintained it is, a certain amount of inherent or natural variability will always exist. This natural variability or "background noise" is the cumulative effect of many small, essentially unavoidable causes. In the framework of statistical quality control, this natural variability is often called a "stable system of chance causes." A process that is operating with only **chance causes of variation** present is said to be **in statistical control**. In other words, the chance causes are an inherent part of the process.

Other kinds of variability may occasionally be present in the output of a process. This variability in key quality characteristics usually arises from three sources: improperly adjusted or controlled machines, operator errors, or defective raw material. Such variability is generally large when compared to the background noise, and it usually represents an unacceptable level of process performance. We refer to these sources of variability that are not part of the chance cause pattern as **assignable causes of variation**. A process that is operating in the presence of assignable causes is said to be an **out-of-control process**.¹

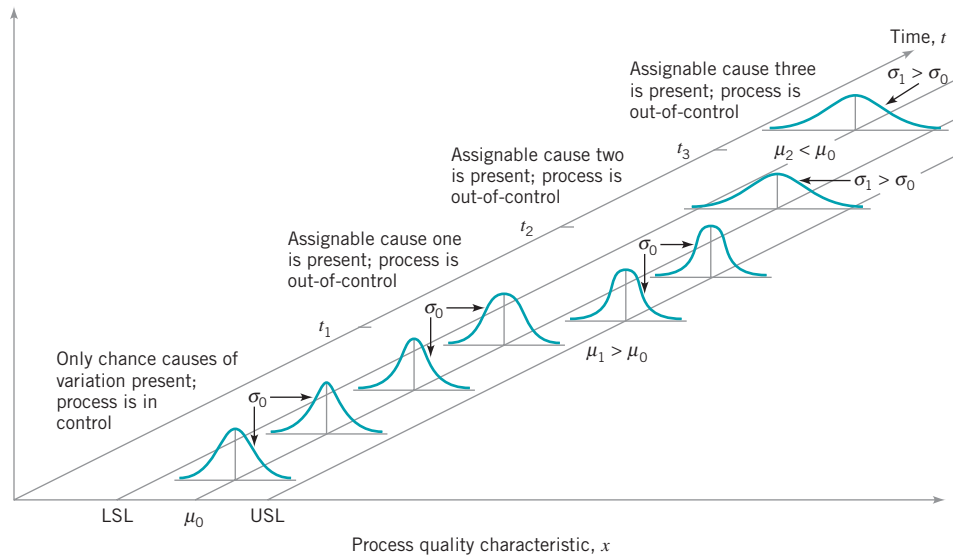
These chance and assignable causes of variation are illustrated in Fig. 5.1. Until time t_1 the process shown in this figure is in control; that is, only chance causes of variation are present. As a result, both the mean and standard deviation of the process are at their in-control values (say, μ_0 and σ_0). At time t_1 an assignable cause occurs. As shown in Fig. 5.1, the effect of this assignable cause is to shift the process mean to a new value $\mu_1 > \mu_0$. At time t_2 another assignable cause occurs, resulting in $\mu = \mu_0$, but now the process standard deviation has shifted to a larger value $\sigma_1 > \sigma_0$. At time t_3 there is another assignable cause present, resulting in both the process mean and standard deviation taking on out-of-control values. From time t_1 forward, the presence of assignable causes has resulted in an out-of-control process.

Processes will often operate in the in-control state for relatively long periods of time. However, no process is truly stable forever, and, eventually, assignable causes will occur, seemingly at random, resulting in a shift to an out-of-control state where a larger proportion of the process output does not conform to requirements. For example, note from Fig. 5.1 that when the process is in control, most of the production will fall between the lower and upper specification limits (LSL and USL, respectively). When the process is out of control, a higher proportion of the process lies outside of these specifications.

A major objective of statistical process control is to quickly detect the occurrence of assignable causes of process shifts so that investigation of the process and corrective action may be undertaken before many nonconforming units are manufactured. The **control chart** is an on-line process-monitoring technique widely used for this purpose. Control charts may also be used to estimate the parameters of a production process, and, through this information, to determine process capability. The control chart may also provide information useful in improving the process. Finally, remember that the eventual goal of statistical process control is the **elimination of variability in the process**. It may not be possible to completely eliminate variability, but the control chart is an effective tool in reducing variability as much as possible.

We now present the statistical concepts that form the basis of control charts. Chapters 6 and 7 develop the details of construction and use of the standard types of control charts.

¹The terminology **chance** and **assignable causes** was developed by Shewhart. Today, some writers use the terminology **common cause** instead of **chance cause** and **special cause** instead of **assignable cause**.

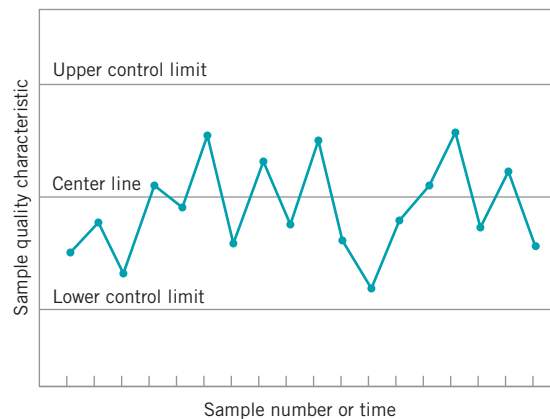


■ FIGURE 5.1 Chance and assignable causes of variation.

5.3 Statistical Basis of the Control Chart

5.3.1 Basic Principles

A typical control chart is shown in Fig. 5.2. The control chart is a graphical display of a quality characteristic that has been measured or computed from a sample versus the sample number or time. The chart contains a **center line** that represents the average value of the quality characteristic corresponding to the in-control state. (That is, only chance causes are present.) Two other horizontal lines, called the **upper control limit (UCL)** and the **lower control limit (LCL)**, are also shown on the chart. These control limits are chosen so that if the process is in control, nearly all of the sample points will fall between them. As long as the points plot within the control limits, the process is assumed to be in control, and no action is necessary. However, a point that plots outside of the control limits is interpreted as evidence that the process is out of control, and investigation and corrective action are required to find and eliminate the assignable cause or causes responsible for this behavior. It is customary to connect the sample points on the control chart with



■ FIGURE 5.2 A typical control chart.

straight-line segments, so that it is easier to visualize how the sequence of points has evolved over time.

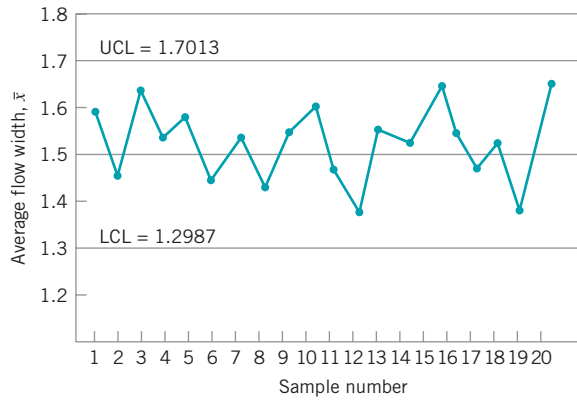
Even if all the points plot inside the control limits, if they behave in a systematic or nonrandom manner, then this could be an indication that the process is out of control. For example, if 18 of the last 20 points plotted above the center line but below the upper control limit and only two of these points plotted below the center line but above the lower control limit, we would be very suspicious that something was wrong. If the process is in control, all the plotted points should have an essentially random pattern. Methods for looking for sequences or nonrandom patterns can be applied to control charts as an aid in detecting out-of-control conditions. Usually, there is a reason why a particular nonrandom pattern appears on a control chart, and if it can be found and eliminated, process performance can be improved. This topic is discussed further in Sections 5.3.5 and 6.2.4.

There is a close connection between control charts and **hypothesis testing**. To illustrate this connection, suppose that the vertical axis in Fig. 5.2 is the sample average $\bar{x}$. Now, if the current value of $\bar{x}$ plots between the control limits, we conclude that the process mean is in control; that is, it is equal to some value μ_0 . On the other hand, if $\bar{x}$ exceeds either control limit, we conclude that the process mean is out of control; that is, it is equal to some value $\mu_1 \neq \mu_0$. In a sense, then, the control chart is a test of the hypothesis that the process is in a state of statistical control. A point plotting within the control limits is equivalent to failing to reject the hypothesis of statistical control, and a point plotting outside the control limits is equivalent to rejecting the hypothesis of statistical control.

The hypothesis testing framework is useful in many ways, but there are some differences in viewpoint between control charts and hypothesis tests. For example, when testing statistical hypotheses, we usually check the validity of assumptions, whereas control charts are used to detect departures from an assumed state of statistical control. In general, we should not worry too much about assumptions such as the form of the distribution or independence when we are applying control charts to a process to reduce variability and achieve statistical control. Furthermore, an assignable cause can result in many different types of shifts in the process parameters. For example, the mean could shift instantaneously to a new value and remain there (this is sometimes called a **sustained shift**); or it could shift abruptly; but the assignable cause could be short-lived and the mean could then return to its nominal or in-control value; or the assignable cause could result in a steady drift or trend in the value of the mean. Only the sustained shift fits nicely within the usual statistical hypothesis testing model.

One place where the hypothesis testing framework is useful is in analyzing the **performance** of a control chart. For example, we may think of the probability of type I error of the control chart (concluding the process is out of control when it is really in control) and the probability of type II error of the control chart (concluding the process is in control when it is really out of control). It is occasionally helpful to use the operating-characteristic curve of a control chart to display its probability of type II error. This would be an indication of the ability of the control chart to detect process shifts of different magnitudes. This can be of value in determining which type of control chart to apply in certain situations. For more discussion of hypothesis testing, the role of statistical theory, and control charts, see Woodall (2000).

To illustrate the preceding ideas, we give an example of a control chart. In semiconductor manufacturing, an important fabrication step is photolithography, in which a light-sensitive photoresist material is applied to the silicon wafer, the circuit pattern is exposed on the resist typically through the use of high-intensity UV light, and the unwanted resist material is removed through a developing process. After the resist pattern is defined, the underlying material is removed by either wet chemical or plasma etching. It is fairly typical to follow development with a hard-bake process to increase resist adherence and etch resistance. An important quality characteristic in hard bake is the flow width of the resist, a measure of how



■ FIGURE 5.3 $\bar{x}$ control chart for flow width.

much it expands due to the baking process. Suppose that flow width can be controlled at a mean of 1.5 microns, and it is known that the standard deviation of flow width is 0.15 microns. A control chart for the average flow width is shown in Fig. 5.3. Every hour, a sample of five wafers is taken, the average flow width ($\bar{x}$) computed, and $\bar{x}$ plotted on the chart. Because this control chart utilizes the sample average $\bar{x}$ to monitor the process mean, it is usually called an $\bar{x}$ control chart. Note that all of the plotted points fall inside the control limits, so the chart indicates that the process is considered to be in statistical control.

To assist in understanding the statistical basis of this control chart, consider how the control limits were determined. The process mean is 1.5 microns, and the process standard deviation is $\sigma = 0.15$ microns. Now if samples of size $n = 5$ are taken, the standard deviation of the sample average $\bar{x}$ is

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{0.15}{\sqrt{5}} = 0.0671$$

Therefore, if the process is in control with a mean flow width of 1.5 microns, then by using the central limit theorem to assume that $\bar{x}$ is approximately normally distributed, we would expect $100(1 - \alpha)\%$ of the sample means $\bar{x}$ to fall between $1.5 + Z_{\alpha/2}(0.0671)$ and $1.5 - Z_{\alpha/2}(0.0671)$. We will arbitrarily choose the constant $Z_{\alpha/2}$ to be 3, so that the upper and lower control limits become

$$UCL = 1.5 + 3(0.0671) = 1.7013$$

and

$$LCL = 1.5 - 3(0.0671) = 1.2987$$

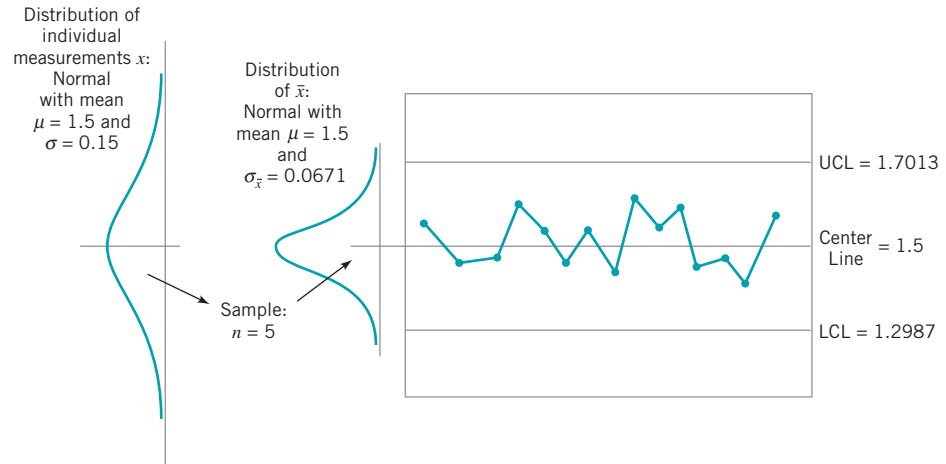
as shown on the control chart. These are typically called **three-sigma control limits**.² The width of the control limits is inversely proportional to the sample size n for a given multiple of sigma. Note that choosing the control limits is equivalent to setting up the critical region for testing the hypothesis

$$H_0: \mu = 1.5$$

$$H_1: \mu \neq 1.5$$

where $\sigma = 0.15$ is known. Essentially, the control chart tests this hypothesis repeatedly at different points in time. The situation is illustrated graphically in Fig. 5.4.

²Note that “sigma” refers to the standard deviation of the statistic plotted on the chart (i.e., $\sigma_{\bar{x}}$), not the standard deviation of the quality characteristic.



■ **FIGURE 5.4** How the control chart works.

We may give a general **model** for a control chart. Let w be a sample statistic that measures some quality characteristic of interest, and suppose that the mean of w is μ_w and the standard deviation of w is σ_w . Then the center line, the upper control limit, and the lower control limit become

$$\begin{aligned} \text{UCL} &= \mu_w + L\sigma_w \\ \text{Center line} &= \mu_w \\ \text{LCL} &= \mu_w - L\sigma_w \end{aligned} \quad (5.1)$$

where L is the “distance” of the control limits from the center line, expressed in standard deviation units. This general theory of control charts was first proposed by Walter A. Shewhart, and control charts developed according to these principles are often called **Shewhart control charts**.

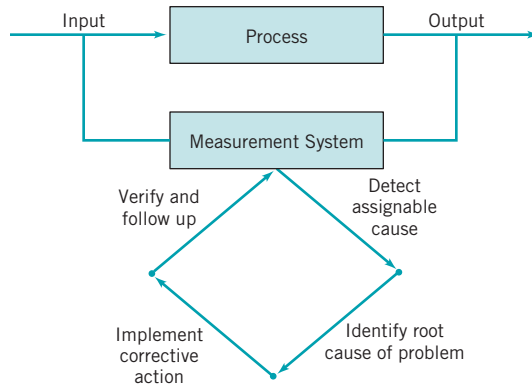
The control chart is a device for describing in a precise manner exactly what is meant by statistical control; as such, it may be used in a variety of ways. In many applications, it is used for on-line process surveillance. That is, sample data are collected and used to construct the control chart, and if the sample values of $\bar{x}$ (say) fall within the control limits and do not exhibit any systematic pattern, we say the process is in control at the level indicated by the chart. Note that we may be interested here in determining *both* whether the past data came from a process that was in control and whether future samples from this process indicate statistical control.

The most important use of a control chart is to **improve** the process. We have found that, generally,

1. Most processes do not operate in a state of statistical control, and
2. Consequently, the routine and attentive use of control charts will identify assignable causes. If these causes can be eliminated from the process, variability will be reduced and the process will be improved.

This process improvement activity using the control chart is illustrated in Fig. 5.5. Note that

3. The control chart will only **detect** assignable causes. Management, operator, and engineering **action** will usually be necessary to eliminate the assignable causes.



■ **FIGURE 5.5** Process improvement using the control chart.

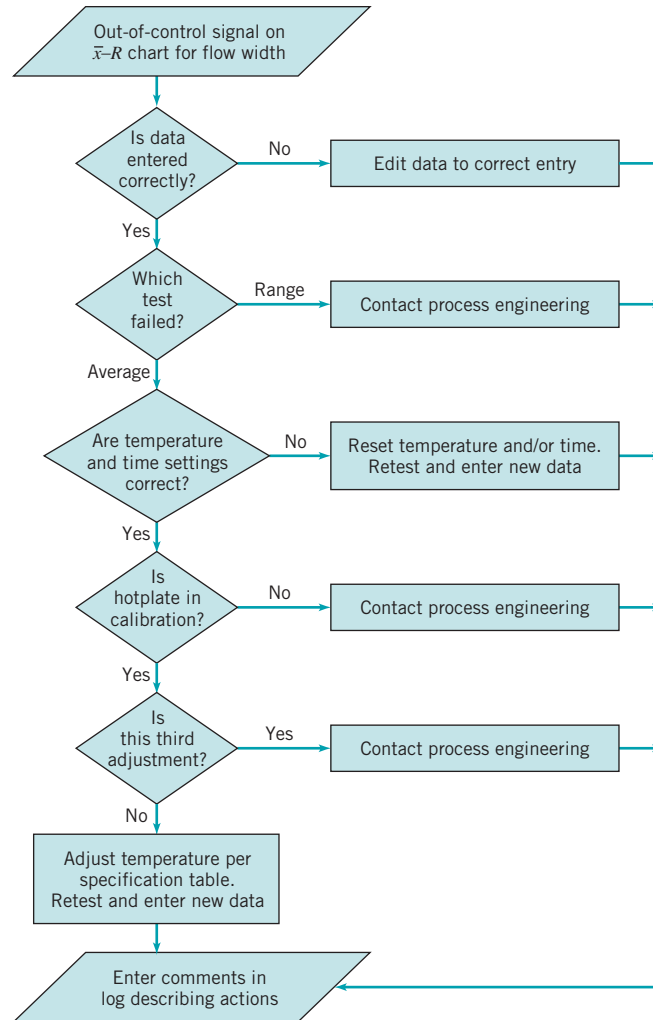
In identifying and eliminating assignable causes, it is important to find the **root cause** of the problem and to attack it. A cosmetic solution will not result in any real, long-term process improvement. Developing an effective system for corrective action is an essential component of an effective SPC implementation.

A very important part of the corrective action process associated with control chart usage is the **out-of-control-action plan (OCAP)**. An OCAP is a flow chart or text-based description of the sequence of activities that must take place following the occurrence of an *activating event*. These are usually out-of-control signals from the control chart. The OCAP consists of *checkpoints*, which are potential assignable causes, and *terminators*, which are actions taken to resolve the out-of-control condition, preferably by eliminating the assignable cause. It is very important that the OCAP specify as complete a set as possible of checkpoints and terminators, and that these be arranged in an order that facilitates process diagnostic activities. Often, analysis of prior failure modes of the process and/or product can be helpful in designing this aspect of the OCAP. Furthermore, an OCAP is a *living document* in the sense that it will be modified over time as more knowledge and understanding of the process is gained. Consequently, when a control chart is introduced, an initial OCAP should accompany it. Control charts without an OCAP are not likely to be useful as a process improvement tool.

The OCAP for the hard-bake process is shown in Fig. 5.6. This process has two controllable variables, temperature and time. In this process, the mean flow width is monitored with an $\bar{x}$ control chart, and the process variability is monitored with a control chart for the range, or an R chart. Notice that if the R chart exhibits an out-of-control signal, operating personnel are directed to contact process engineering immediately. If the $\bar{x}$ control chart exhibits an out-of-control signal, operators are directed to check process settings and calibration and then make adjustments to temperature in an effort to bring the process back into a state of control. If these adjustments are unsuccessful, process engineering personnel are contacted.

We may also use the control chart as an **estimating device**. That is, from a control chart that exhibits statistical control, we may estimate certain process parameters, such as the mean, standard deviation, fraction nonconforming or fallout, and so forth. These estimates may then be used to determine the **capability** of the process to produce acceptable products. Such **process-capability studies** have considerable impact on many management decision problems that occur over the product cycle, including make or buy decisions, plant and process improvements that reduce process variability, and contractual agreements with customers or vendors regarding product quality.

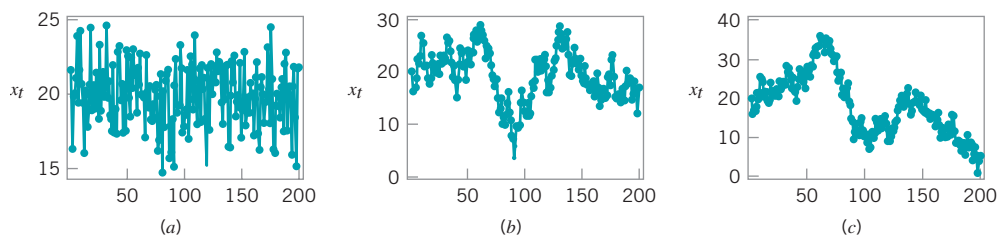
Control charts may be classified into two general types. If the quality characteristic can be measured and expressed as a number on some continuous scale of measurement, it is usually called a **variable**. In such cases, it is convenient to describe the quality characteristic with a measure of central tendency and a measure of variability. Control charts for central tendency



■ **FIGURE 5.6** The out-of-control-action plan (OCAP) for the hard-bake process.

and variability are collectively called **variables control charts**. The $\bar{x}$ chart is the most widely used chart for controlling central tendency, whereas charts based on either the sample range or the sample standard deviation are used to control process variability. Control charts for variables are discussed in Chapter 6. Many quality characteristics are not measured on a continuous scale or even a quantitative scale. In these cases, we may judge each unit of product as either conforming or nonconforming on the basis of whether or not it possesses certain attributes, or we may count the number of nonconformities (defects) appearing on a unit of product. Control charts for such quality characteristics are called **attributes control charts** and are discussed in Chapter 7.

An important factor in control chart use is the **design of the control chart**. By this we mean the selection of the sample size, control limits, and frequency of sampling. For example, in the $\bar{x}$ chart of Fig. 5.3, we specified a sample size of five measurements, three-sigma control limits, and the sampling frequency to be every hour. In most quality-control problems, it is customary to design the control chart using primarily statistical considerations. For example, we know that increasing the sample size will decrease the probability of type II error, thus enhancing the chart's ability to detect an out-of-control state, and so forth. The use of statistical criteria such as these along with industrial experience has led to general guidelines and



■ **FIGURE 5.7** Data from three different processes. (a) Stationary and uncorrelated (white noise). (b) Stationary and autocorrelated. (c) Nonstationary.

procedures for designing control charts. These procedures usually consider cost factors only in an implicit manner. Recently, however, we have begun to examine control chart design from an **economic** point of view, considering explicitly the cost of sampling, losses from allowing defective product to be produced, and the costs of investigating out-of-control signals that are really false alarms.

Another important consideration in control chart usage is the **type of variability** exhibited by the process. Fig. 5.7 presents data from three different processes. Figures 5.7a and 5.7b illustrate **stationary behavior**. By this we mean that the process data vary around a fixed mean in a stable or predictable manner. This is the type of behavior that Shewhart implied was produced by an **in-control process**.

Even a cursory examination of Figs. 5.7a and 5.7b reveals some important differences. The data in Fig. 5.7a are **uncorrelated**; that is, the observations give the appearance of having been drawn at random from a stable population, perhaps a normal distribution. This type of data is referred to by time series analysts as **white noise**. (Time-series analysis is a field of statistics devoted exclusively to studying and modeling time-oriented data.) In this type of process, the order in which the data occur does not tell us much that is useful to analyze the process. In other words, the past values of the data are of no help in predicting any of the future values.

Figure 5.7b illustrates stationary but **autocorrelated** process data. Notice that successive observations in these data are **dependent**; that is, a value above the mean tends to be followed by another value above the mean, whereas a value below the mean is usually followed by another such value. This produces a data series that has a tendency to move in moderately long “runs” on either side of the mean.

Figure 5.7c illustrates **nonstationary** variation. This type of process data occurs frequently in the chemical and process industries. Note that the process is very unstable in that it drifts or “wanders about” without any sense of a stable or fixed mean. In many industrial settings, we stabilize this type of behavior by using **engineering process control**, (such as **feedback control**). This approach to process control is required when there are factors that affect the process that cannot be stabilized, such as environmental variables or properties of raw materials. When the control scheme is effective, the process output will *not* look like Fig. 5.7c, but will resemble either Fig. 5.7a or 5.7b.

Shewhart control charts are most effective when the in-control process data look like Fig. 5.7a. By this we mean that the charts can be designed so that their performance is predictable and reasonable to the user, and that they are effective in reliably detecting out-of-control conditions. Most of our discussion of control charts in this chapter and in Chapters 6 and 7 will assume that the in-control process data are stationary and uncorrelated.

With some modifications, Shewhart control charts and other types of control charts can be applied to autocorrelated data. We discuss this in more detail in Part IV of the book. We also discuss feedback control and the use of SPC in systems where feedback control is employed in Part IV.

Control charts have had a long history of use in U.S. industries and in many offshore industries as well. There are at least five reasons for their popularity.

1. **Control charts are a proven technique for improving productivity.** A successful control chart program will reduce scrap and rework, which are the primary productivity killers in *any* operation. If you reduce scrap and rework, then productivity increases, cost decreases, and production capacity (measured in the number of *good* parts per hour) increases.
2. **Control charts are effective in defect prevention.** The control chart helps keep the process in control, which is consistent with the “do it right the first time” philosophy. It is never cheaper to sort out “good” units from “bad” units later on than it is to build it right initially. If you do not have effective process control, you are paying someone to make a nonconforming product.
3. **Control charts prevent unnecessary process adjustment.** A control chart can distinguish between background noise and abnormal variation; no other device including a human operator is as effective in making this distinction. If process operators adjust the process based on periodic tests unrelated to a control chart program, they will often overreact to the background noise and make unneeded adjustments. Such unnecessary adjustments can actually result in a deterioration of process performance. In other words, the control chart is consistent with the “if it isn’t broken, don’t fix it” philosophy.
4. **Control charts provide diagnostic information.** Frequently, the pattern of points on the control chart will contain information of diagnostic value to an experienced operator or engineer. This information allows the implementation of a change in the process that improves its performance.
5. **Control charts provide information about process capability.** The control chart provides information about the value of important process parameters and their stability over time. This allows an estimate of process capability to be made. This information is of tremendous use to product and process designers.

Control charts are among the most important management control tools; they are as important as cost controls and material controls. Modern computer technology has made it easy to implement control charts in *any* type of process, as data collection and analysis can be performed on a microcomputer or a local area network terminal in real-time, on-line at the work center. Some additional guidelines for implementing a control chart program are given at the end of Chapter 7.

5.3.2 Choice of Control Limits

Specifying the **control limits** is one of the critical decisions that must be made in designing a control chart. By moving the control limits farther from the center line, we decrease the risk of a type I error—that is, the risk of a point falling beyond the control limits, indicating an out-of-control condition when no assignable cause is present. However, widening the control limits will also increase the risk of a type II error—that is, the risk of a point falling between the control limits when the process is really out of control. If we move the control limits closer to the center line, the opposite effect is obtained: The risk of type I error is increased, while the risk of type II error is decreased.

For the $\bar{x}$ chart shown in Fig. 5.3, where three-sigma control limits were used, if we assume that the flow width is normally distributed, we find from the standard normal table that the probability of type I error is 0.0027. That is, an incorrect out-of-control signal or false alarm will be generated in only 27 out of 10,000 points. Furthermore, the probability that a point taken when the process is in control will exceed the three-sigma limits in one direction

only is 0.00135. Instead of specifying the control limit as a multiple of the standard deviation of $\bar{x}$, we could have directly chosen the type I error probability and calculated the corresponding control limit. For example, if we specified a 0.001 type I error probability in one direction, then the appropriate multiple of the standard deviation would be 3.09. The control limits for the $\bar{x}$ chart would then be

$$\text{UCL} = 1.5 + 3.09(0.0671) = 1.7073$$

$$\text{LCL} = 1.5 - 3.09(0.0671) = 1.2927$$

These control limits are usually called **0.001 probability limits**, although they should logically be called 0.002 probability limits, because the total risk of making a type I error is 0.002. There is only a slight difference between the two limits.

Regardless of the distribution of the quality characteristic, it is standard practice in the United States to determine the control limits as a multiple of the standard deviation of the statistic plotted on the chart. The multiple usually chosen is three; hence, three-sigma limits are customarily employed on control charts, regardless of the type of chart employed. In the United Kingdom and parts of Western Europe, probability limits are used, with the standard probability level in each direction being 0.001.

We typically justify the use of three-sigma control limits on the basis that they give good results in practice. Moreover, in many cases, the true distribution of the quality characteristic is not known well enough to compute exact probability limits. If the distribution of the quality characteristic is reasonably approximated by the normal distribution, then there will be little difference between three-sigma and 0.001 probability limits.

Two Limits on Control Charts. Some analysts suggest using two sets of limits on control charts, such as those shown in Fig. 5.8. The outer limits—say, at three-sigma—are the usual **action limits**; that is, when a point plots outside of this limit, a search for an assignable cause is made and corrective action is taken if necessary. The inner limits, usually at two-sigma, are called **warning limits**. In Fig. 5.8, we have shown the three-sigma upper and lower control limits for the $\bar{x}$ chart for flow width. The upper and lower warning limits are located at

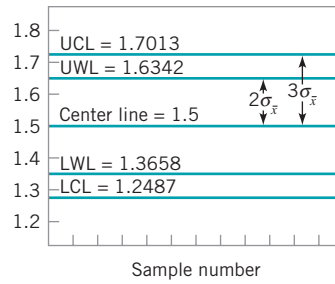
$$\text{UWL} = 1.5 + 2(0.0671) = 1.6342$$

$$\text{LWL} = 1.5 - 2(0.0671) = 1.3658$$

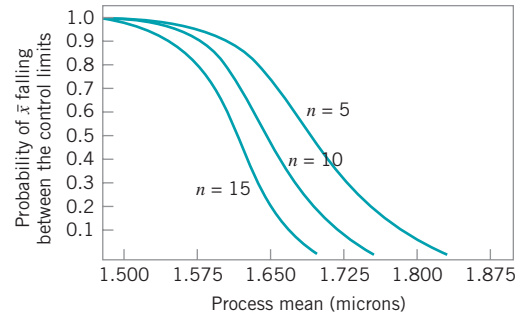
When probability limits are used, the action limits are generally 0.001 limits and the warning limits are 0.025 limits.

If one or more points fall between the warning limits and the control limits, or very close to the warning limit, we should be suspicious that the process may not be operating properly. One possible action to take when this occurs is to increase the sampling frequency and/or the sample size so that more information about the process can be obtained quickly. Process control schemes that change the sample size and/or the sampling frequency depending on the position of the current sample value are called **adaptive** or **variable sampling interval** (or **variable sample size**, etc.) schemes. These techniques have been used in practice for many years and have recently been studied extensively by researchers in the field. We will discuss this technique again in Part IV of this book.

The use of warning limits can increase the **sensitivity** of the control chart; that is, it can allow the control chart to signal a shift in the process more quickly. One of their disadvantages is that they may be confusing to operating personnel. This is not usually a serious objection, however, and many practitioners use warning limits routinely on control charts. A more serious objection is that although the use of warning limits can improve the sensitivity of the chart, they also result in an increased risk of false alarms. We will discuss the use of sensitizing rules (such as warning limits) more thoroughly in Section 5.3.6.



■ **FIGURE 5.8** An $\bar{x}$ chart with two-sigma and three-sigma warning limits.



■ **FIGURE 5.9** Operating-characteristic curves for an $\bar{x}$ chart.

5.3.3 Sample Size and Sampling Frequency

In designing a control chart, we must specify both the **sample size** and the **frequency of sampling**. In general, larger samples will make it easier to detect small shifts in the process. This is demonstrated in Fig. 5.9, where we have plotted the operating-characteristic curve for the $\bar{x}$ chart in Fig. 5.3 for various sample sizes. Note that the probability of detecting a shift from 1.500 microns to 1.650 microns (for example) increases as the sample size n increases. When choosing the sample size, we must keep in mind the size of the shift that we are trying to detect. If the process shift is relatively large, then we use smaller sample sizes than those that would be employed if the shift of interest were relatively small.

We must also determine the frequency of sampling. The most desirable situation from the point of view of detecting shifts would be to take large samples very frequently; however, this is usually not economically feasible. The general problem is one of **allocating sampling effort**. That is, either we take small samples at short intervals or larger samples at longer intervals. Current industry practice tends to favor smaller, more frequent samples, particularly in high-volume manufacturing processes, or where a great many types of assignable causes can occur. Furthermore, as automatic sensing and measurement technology develops, it is becoming possible to greatly increase sampling frequencies. Ultimately, every unit can be tested as it is manufactured. Automatic measurement systems and microcomputers with SPC software applied at the work center for real-time, on-line process control is an effective way to apply statistical process control.

Another way to evaluate the decisions regarding sample size and sampling frequency is through the **average run length (ARL)** of the control chart. Essentially, the ARL is the average number of points that must be plotted before a point indicates an out-of-control condition. If the process observations are uncorrelated, then for any Shewhart control chart, the ARL can be calculated easily from

$$ARL = \frac{1}{p} \quad (5.2)$$

where p is the probability that any point exceeds the control limits. This equation can be used to evaluate the performance of the control chart.

To illustrate, for the $\bar{x}$ chart with three-sigma limits, $p = 0.0027$ is the probability that a single point falls outside the limits when the process is in control. Therefore, the average run length of the $\bar{x}$ chart when the process is in control (called ARL_0) is

$$ARL_0 = \frac{1}{p} = \frac{1}{0.0027} = 370$$

That is, even if the process remains in control, an out-of-control signal will be generated every 370 samples, on the average.

The use of average run lengths to describe the performance of control charts has been subjected to criticism in recent years. The reasons for this arise because the distribution of run length for a Shewhart control chart is a geometric distribution (refer to Section 3.2.4). Consequently, there are two concerns with ARL: (1) the standard deviation of the run length is very large, and (2) the geometric distribution is very skewed, so the mean of the distribution (the ARL) is not necessarily a very typical value of the run length.

For example, consider the Shewhart $\bar{x}$ control chart with three-sigma limits. When the process is in control, we have noted that $p = 0.0027$ and the in-control ARL_0 is $ARL_0 = 1/p = 1/0.0027 = 370$. This is the mean of the geometric distribution. Now the standard deviation of the geometric distribution is

$$\sqrt{(1-p)/p} = \sqrt{(1-0.0027)/0.0027} \cong 370$$

That is, the standard deviation of the geometric distribution in this case is approximately equal to its mean. As a result, the actual ARL_0 observed in practice for the Shewhart $\bar{x}$ control chart will likely vary considerably. Furthermore, for the geometric distribution with $p = 0.0027$, the tenth and fiftieth percentiles of the distribution are 38 and 256, respectively. This means that approximately 10% of the time the in-control run length will be less than or equal to 38 samples and 50% of the time it will be less than or equal to 256 samples. This occurs because the geometric distribution with $p = 0.0027$ is quite skewed to the right. For this reason, some analysts like to report percentiles of the run length distribution instead of just the ARL.

It is also occasionally convenient to express the performance of the control chart in terms of its **average time to signal (ATS)**. If samples are taken at fixed intervals of time that are h hours apart, then

$$ATS = ARLh \quad (5.3)$$

Consider the hard-bake process discussed earlier, and suppose we are sampling every hour. Equation (5.3) indicates that we will have a **false alarm** about every 370 hours on the average.

Now consider how the control chart performs in detecting shifts in the mean. Suppose we are using a sample size of $n = 5$ and that when the process goes out of control the mean shifts to 1.725 microns. From the operating characteristic curve in Fig. 5.9 we find that if the process mean is 1.725 microns, the probability of $\bar{x}$ falling between the control limits is approximately 0.35. Therefore, p in equation (5.2) is 0.35, and the out-of-control ARL (called ARL_1) is

$$ARL_1 = \frac{1}{p} = \frac{1}{0.35} = 2.86$$

That is, the control chart will require 2.86 samples to detect the process shift, on the average, and since the time interval between samples is $h = 1$ hour, the average time required to detect this shift is

$$ATS = ARL_1 h = 2.86 (1) = 2.86 \text{ hours}$$

Suppose that this is unacceptable, because production of wafers with mean flow width of 1.725 microns results in excessive scrap costs and can result in further upstream manufacturing problems. How can we reduce the time needed to detect the out-of-control condition? One method is to sample more frequently. For example, if we sample every half hour, then the average time to signal for this scheme is $ATS = ARL_1 h = 2.86(\frac{1}{2}) = 1.43$; that is, only

1.43 hours will elapse (on the average) between the shift and its detection. The second possibility is to increase the sample size. For example, if we use $n = 10$, then Fig. 5.9 shows that the probability of $\bar{x}$ falling between the control limits when the process mean is 1.725 microns is approximately 0.1, so that $p = 0.9$, and from equation (5.2) the out-of-control ARL or ARL_1 is

$$ARL_1 = \frac{1}{p} = \frac{1}{0.9} = 1.11$$

and, if we sample every hour, the average time to signal is

$$ATS = ARL_1 h = 1.11(1) = 1.11 \text{ hours}$$

Thus, the larger sample size would allow the shift to be detected more quickly than with the smaller one.

To answer the question of sampling frequency more precisely, we must take several factors into account, including the cost of sampling, the losses associated with allowing the process to operate out of control, the rate of production, and the probabilities with which various types of process shifts occur. We discuss various methods for selecting an appropriate sample size and sampling frequency for a control chart in the next four chapters.

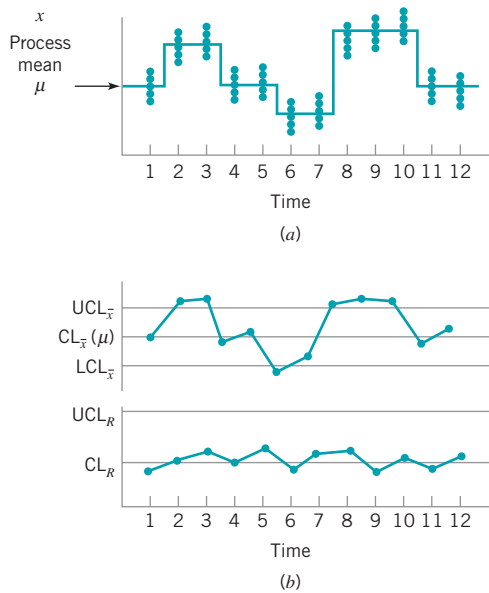
5.3.4 Rational Subgroups

A fundamental idea in the use of control charts is the collection of sample data according to what Shewhart called the **rational subgroup** concept. To illustrate this concept, suppose that we are using an $\bar{x}$ control chart to detect changes in the process mean. Then the rational subgroup concept means that subgroups or samples should be selected so that if assignable causes are present, the chance for differences *between* subgroups will be maximized, while the chance for differences due to these assignable causes *within* a subgroup will be minimized.

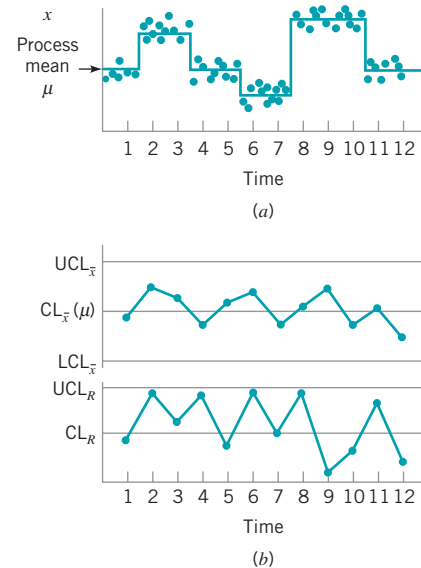
When control charts are applied to production processes, the time order of production is a logical basis for rational subgrouping. Even though time order is preserved, it is still possible to form subgroups erroneously. If some of the observations in the sample are taken at the end of one shift and the remaining observations are taken at the start of the next shift, then any differences between shifts might not be detected. Time order is frequently a good basis for forming subgroups because it allows us to detect assignable causes that occur over time.

Two general approaches to constructing rational subgroups are used. In the first approach, each sample consists of units that were produced at the same time (or as closely together as possible). Ideally, we would like to take **consecutive** units of production. This approach is used when the primary purpose of the control chart is to detect process shifts. It minimizes the chance of variability due to assignable causes *within* a sample, and it maximizes the chance of variability *between* samples if assignable causes are present. It also provides a better estimate of the standard deviation of the process in the case of variables control charts. This approach to rational subgrouping essentially gives a snapshot of the process at each point in time where a sample is collected.

Figure 5.10 illustrates this type of sampling strategy. In Fig. 5.10a we show a process for which the mean experiences a series of sustained shifts, and the corresponding observations obtained from this process at the points in time along the horizontal axis, assuming that five consecutive units are selected. Figure 5.10b shows the $\bar{x}$ **control chart** and an **R chart** (or **range chart**) for these data. The center line and control limits on the **R chart** are constructed using the range of each sample in the upper part of the figure (details will be given in Chapter 6). Note that although the process mean is shifting, the process variability is stable. Furthermore, the within-sample measure of variability is used to construct the control



■ **FIGURE 5.10** The snapshot approach to rational subgroups. (a) Behavior of the process mean. (b) Corresponding $\bar{x}$ and R control charts.



■ **FIGURE 5.11** The random sample approach to rational subgroups. (a) Behavior of the process mean. (b) Corresponding $\bar{x}$ and R control charts.

limits on the $\bar{x}$ chart. Note that the $\bar{x}$ chart in Fig. 5.10b has points out of control corresponding to the shifts in the process mean.

In the second approach, each sample consists of units of product that are representative of *all* units that have been produced since the last sample was taken. Essentially, each subgroup is a **random sample of all process output over the sampling interval**. This method of rational subgrouping is often used when the control chart is employed to make decisions about the acceptance of all units of product that have been produced since the last sample. In fact, if the process shifts to an out-of-control state and then back in control again *between* samples, it is sometimes argued that the snapshot method of rational subgrouping will be ineffective against these types of shifts, and so the random sample method must be used.

When the rational subgroup is a random sample of all units produced over the sampling interval, considerable care must be taken in interpreting the control charts. If the process mean drifts between several levels during the interval between samples, this may cause the range of the observations within the sample to be relatively large, resulting in wider limits on the $\bar{x}$ chart. This scenario is illustrated in Fig. 5.11. In fact, **we can often make any process appear to be in statistical control just by stretching out the interval between observations in the sample**. It is also possible for shifts in the process average to cause points on a control chart for the range or standard deviation to plot out of control, even though there has been no shift in process variability.

There are other bases for forming rational subgroups. For example, suppose a process consists of several machines that pool their output into a common stream. If we sample from this common stream of output, it will be very difficult to detect whether any of the machines are out of control. A logical approach to rational subgrouping here is to apply control chart techniques to the output for each individual machine. Sometimes this concept needs to be applied to different heads on the same machine, different work stations, different operators, and so forth. In many situations, the rational subgroup will consist of a single observation. This situation occurs frequently in the chemical and process industries where the quality

characteristic of the product changes relatively slowly and samples taken very close together in time are virtually identical, apart from measurement or analytical error.

The rational subgroup concept is very important. The proper selection of samples requires careful consideration of the process, with the objective of obtaining as much useful information as possible from the control chart analysis.

5.3.5 Analysis of Patterns on Control Charts

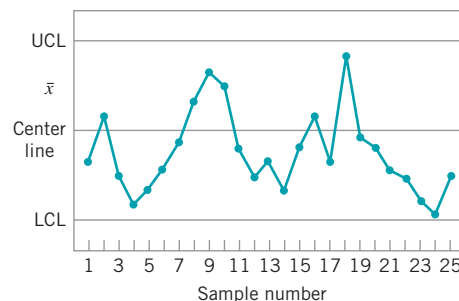
Patterns on control charts must be assessed. A control chart may indicate an out-of-control condition when one or more points fall beyond the control limits or when the plotted points exhibit some nonrandom pattern of behavior. For example, consider the $\bar{x}$ chart shown in Fig. 5.12. Although all 25 points fall within the control limits, the points do not indicate statistical control because their pattern is very nonrandom in appearance. Specifically, we note that 19 of 25 points plot below the center line, while only 6 of them plot above. If the points truly are random, we should expect a more even distribution above and below the center line. We also observe that following the fourth point, five points in a row increase in magnitude. This arrangement of points is called a **run**. Since the observations are increasing, we could call this a *run up*. Similarly, a sequence of decreasing points is called a *run down*. This control chart has an unusually long run up (beginning with the fourth point) and an unusually long run down (beginning with the eighteenth point).

In general, we define a run as a sequence of observations of the same type. In addition to runs up and runs down, we could define the types of observations as those above and below the center line, respectively, so that two points in a row above the center line would be a run of length 2.

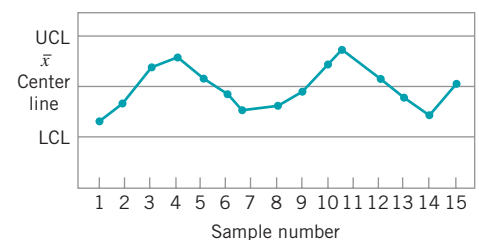
A run of length 8 or more points has a very low probability of occurrence in a random sample of points. Consequently, any type of run of length 8 or more is often taken as a signal of an out-of-control condition. For example, eight consecutive points on one side of the center line may indicate that the process is out of control.

Although runs are an important measure of nonrandom behavior on a control chart, other types of patterns may also indicate an out-of-control condition. For example, consider the $\bar{x}$ chart in Fig. 5.13. Note that the plotted sample averages exhibit a cyclic behavior, yet they all fall within the control limits. Such a pattern may indicate a problem with the process such as operator fatigue, raw material deliveries, heat or stress buildup, and so forth. Although the process is not really out of control, the yield may be improved by elimination or reduction of the sources of variability causing this cyclic behavior (see Fig. 5.14).

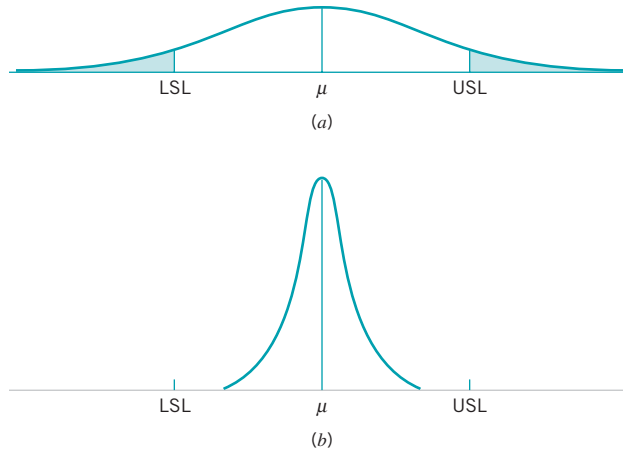
The problem is one of **pattern recognition**—that is, recognizing systematic or nonrandom patterns on the control chart and identifying the reason for this behavior. The ability to interpret a particular pattern in terms of assignable causes requires experience and knowledge



■ **FIGURE 5.12** An $\bar{x}$ control chart.



■ **FIGURE 5.13** An $\bar{x}$ chart with a cyclic pattern.



■ **FIGURE 5.14** (a) Variability with the cyclic pattern. (b) Variability with the cyclic pattern eliminated.

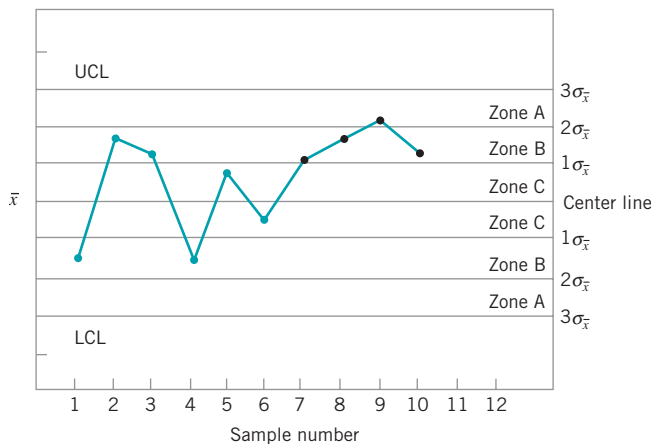
of the process. That is, we must not only know the statistical principles of control charts, but we must also have a good understanding of the process. We discuss the interpretation of patterns on control charts in more detail in Chapter 6.

The Western Electric Handbook (1956) suggests a set of decision rules for detecting nonrandom patterns on control charts. Specifically, it suggests concluding that the process is out of control if either

1. One point plots outside the three-sigma control limits,
2. Two out of three consecutive points plot beyond the two-sigma warning limits,
3. Four out of five consecutive points plot at a distance of one-sigma or beyond from the center line, or
4. Eight consecutive points plot on one side of the center line.

Those rules apply to one side of the center line at a time. Therefore, a point above the *upper* warning limit followed immediately by a point below the *lower* warning limit would not signal an out-of-control alarm. These are often used in practice for enhancing the sensitivity of control charts. That is, the use of these rules can allow smaller process shifts to be detected more quickly than would be the case if our only criterion was the usual three-sigma control limit violation.

Figure 5.15 shows an $\bar{x}$ control chart with the one-sigma, two-sigma, and three-sigma limits used in the Western Electric procedure. Note that these limits partition the control chart



■ **FIGURE 5.15** The Western Electric or zone rules, with the last four points showing a violation of rule 3.

■ **TABLE 5.1**
Some Sensitizing Rules for Shewhart Control Charts

Standard Action Signal:	1. One or more points outside of the control limits. 2. Two of three consecutive points outside the two-sigma warning limits but still inside the control limits. 3. Four of five consecutive points beyond the one-sigma limits. 4. A run of eight consecutive points on one side of the center line. 5. Six points in a row steadily increasing or decreasing. 6. Fifteen points in a row in zone C (both above and below the center line). 7. Fourteen points in a row alternating up and down. 8. Eight points in a row on both sides of the center line with none in zone C. 9. An unusual or nonrandom pattern in the data. 10. One or more points near a warning or control limit.	Western Electric Rules
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into three zones (A, B, and C) on each side of the center line. Consequently, the Western Electric rules are sometimes called the **zone rules** for control charts. Note that the last four points fall in zone B or beyond. Thus, since four of five consecutive points exceed the one-sigma limit, the Western Electric procedure will conclude that the pattern is nonrandom and the process is out of control.

5.3.6 Discussion of Sensitizing Rules for Control Charts

As may be gathered from earlier sections, several criteria may be applied simultaneously to a control chart to determine whether the process is out of control. The basic criterion is one or more points outside of the control limits. The supplementary criteria are sometimes used to increase the **sensitivity** of the control charts to a small process shift so that we may respond more quickly to the assignable cause. Some of the **sensitizing rules for control charts** that are widely used in practice are shown in Table 5.1. For a good discussion of some of these rules, see Nelson (1984). Frequently, we will inspect the control chart and conclude that the process is out of control if any one or more of the criteria in Table 5.1 are met.

When several of these sensitizing rules are applied simultaneously, we often use a **graduated response** to out-of-control signals. For example, if a point exceeded a control limit, we would immediately begin to search for the assignable cause, but if one or two consecutive points exceeded only the two-sigma warning limit, we might increase the frequency of sampling from every hour to say, every 10 minutes. This **adaptive sampling response** might not be as severe as a complete search for an assignable cause, but if the process were really out of control, it would give us a high probability of detecting this situation more quickly than we would by maintaining the longer sampling interval.

In general, care should be exercised when using several decision rules simultaneously. Suppose that the analyst uses k decision rules and that criterion i has type I error probability α_i . Then the overall type I error or false alarm probability for the decision based on all k tests is

$$\alpha = 1 - \prod_{i=1}^k (1 - \alpha_i) \quad (5.4)$$

provided that all k decision rules are independent. However, the independence assumption is not valid with the usual sensitizing rules. Furthermore, the value of α_i is not always clearly defined for the sensitizing rules, because these rules involve several observations.

Champ and Woodall (1987) investigated the average run length performance for the Shewhart control chart with various sensitizing rules. They found that the use of these rules does improve the ability of the control chart to detect smaller shifts, but the in-control average run length can be substantially degraded. For example, assuming independent process data and using a Shewhart control chart with the Western Electric rules results in an in-control ARL of 91.25, in contrast to 370 for the Shewhart control chart alone.

Some of the individual Western Electric rules are particularly troublesome. An illustration is the rule of several (usually seven or eight) consecutive points which either increase or decrease. This rule is very ineffective in detecting a trend, the situation for which it was designed. It does, however, greatly increase the false alarm rate. See Davis and Woodall (1988) for more details.

5.3.7 Phase I and Phase II of Control Chart Application

Standard control chart usage involves **phase I and phase II applications**, with two different and distinct objectives. In phase I, a set of process data is gathered and analyzed all at once in a **retrospective** analysis, constructing **trial control limits** to determine if the process has been in control over the period of time where the data were collected, and to see if reliable control limits can be established to monitor future production. This is typically the first thing that is done when control charts are applied to any process. Control charts in phase I primarily assist operating personnel in bringing the process into a state of statistical control. Phase II beings after we have a “clean” set of process data gathered under stable conditions and representative of in-control process performance. In phase II, we use the control chart to **monitor** the process by comparing the sample statistic for each successive sample as it is drawn from the process to the control limits.

Thus, in phase I we are comparing a collection of, say, m points to a set of control limits computed from those points. Typically $m = 20$ or 25 subgroups are used in phase I. It is fairly typical in phase I to assume that the process is initially out of control, so the objective of the analyst is to bring the process into a state of statistical control. Control limits are calculated based on the m subgroups and the data plotted on the control charts. Points that are outside the control limits are investigated, looking for potential assignable causes. Any assignable causes that are identified are worked on by engineering and operating personnel in an effort to eliminate them. Points outside the control limits are then excluded and a new set of revised control limits calculated. Then new data are collected and compared to these revised limits. Sometimes this type of analysis will require several cycles in which the control chart is employed, assignable causes are detected and corrected, revised control limits are calculated, and the out-of-control action plan is updated and expanded. Eventually the process is stabilized, and a clean set of data that represents in-control process performance is obtained for use in phase II.

Generally, Shewhart control charts are very effective in phase I because they are easy to construct and interpret, and because they are effective in detecting both large, sustained shifts in the process parameters and outliers (single excursions that may have resulted from assignable causes of short duration), measurement errors, data recording and/or transmission errors, and the like. Furthermore, patterns on Shewhart control charts often are easy to interpret and have direct physical meaning. The sensitizing rules discussed in the previous sections are also easy to apply to Shewhart charts. (This is an optional feature in most SPC software.) The types of assignable causes that usually occur in phase I result in fairly large process shifts—exactly

the scenario in which the Shewhart control chart is most effective. Average run length is not usually a reasonable performance measure for phase I; we are typically more interested in the probability that an assignable cause will be detected than in the occurrence of false alarms. For good discussions of phase I control chart usage and related matters, see the papers by Woodall (2000), Borror and Champ (2001), Boyles (2000), and Champ and Chou (2003), and the standard ANSI/ASQC B1–133–1996 Quality Control Chart Methodologies (this can be downloaded at e-standards.asq.org).

In phase II, we usually assume that the process is reasonably stable. Often, the assignable causes that occur in phase II result in smaller process shifts, because (it is hoped) most of the really ugly sources of variability have been systematically removed during phase I. Our emphasis is now on **process monitoring**, not on bringing an unruly process into control. Average run length is a valid basis for evaluating the performance of a control chart in phase II. Shewhart control charts are much less likely to be effective in phase II because they are not very sensitive to small to moderate size process shifts; that is, their ARL performance is relatively poor. Attempts to solve this problem by employing sensitizing rules such as those discussed in the previous section are likely to be unsatisfactory, because the use of these supplemental sensitizing rules increases the false alarm rate of the Shewhart control chart. [Recall the discussion of the Champ and Woodall (1987) paper in the previous section.] The routine use of sensitizing rules to detect small shifts or to react more quickly to assignable causes in phase II should be discouraged. The cumulative sum and EWMA control charts discussed in Chapter 9 are much more likely to be effective in phase II.

5.4 The Rest of the Magnificent Seven

Although the control chart is a very powerful problem-solving and process-improvement tool, it is most effective when its use is fully integrated into a comprehensive SPC program. The seven major SPC problem-solving tools should be widely taught throughout the organization and used routinely to identify improvement opportunities and to assist in reducing variability and eliminating waste. They can be used in several ways throughout the DMAIC problem-solving process. The magnificent seven, introduced in Section 5.1, are listed again here for convenience:

1. Histogram or stem-and-leaf plot
2. Check sheet
3. Pareto chart
4. Cause-and-effect diagram
5. Defect concentration diagram
6. Scatter diagram
7. Control chart

We introduced the histogram and the stem-and-leaf plot in Chapter 3, and discussed the control chart in Section 5.3. In this section, we illustrate the rest of the tools.

Check Sheet. In the early stages of process improvement, it will often become necessary to collect either historical or current operating data about the process under investigation. This is a common activity in the measure step of DMAIC. A **check sheet** can be very useful in this data collection activity. The check sheet shown in Fig. 5.16 was developed by an aerospace firm engineer who was investigating defects that occurred on one of the firm's tanks. The engineer designed the check sheet to help summarize all the